



USBD2.0 PROGRAMMING GUIDE

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1. USB Device Controller

1.1. General Description

In GPA7XXXA, there're 2 sets of USB device controller that support USB2.0 High speed and full speed. Every controller contains 11 Endpoints, and there're 7 types of endpoint, including Control Pipe, bulk in, bulk out, interrupt in, isochronous in and isochronous out.

The abbreviations used in this chapter are listed in the following table.

USB	Universal Serial Bus
HS	High-Speed
FS	Full-Speed
UTMI	USB 2.0 Transceiver Macrocell Interface
PHY	Physical Layer
DMA	Direct Memory Access
EP	EndPoint
UDC	USB Device Controller
UVC	USB Video Class

1.2. Features

- Support USB2.0 High Speed and Full Speed.
- UTMI Interface to USB PHY
- Support four transfer types and 11 endpoints.
 - Control IN/OUT: Endpoint 0
 - BULK IN/OUT
 - IN: Endpoint 1, Endpoint 8
 - OUT: Endpoint 2, Endpoint 9
 - Isochronous IN
 - Video: Endpoint 5 (Maximum Bandwidth: 1024x8x3 bytes/ms)
 - Audio: Endpoint 7 (Maximum Bandwidth: 256x8 bytes/ms)
 - Interrupt IN: Endpoint 3, Endpoint 4 and Endpoint 6
 - Audio Isochronous OUT: Endpoint 12 (Maximum Bandwidth:1024x8 bytes/ms)

1.3. Block Diagram for All Endpoints

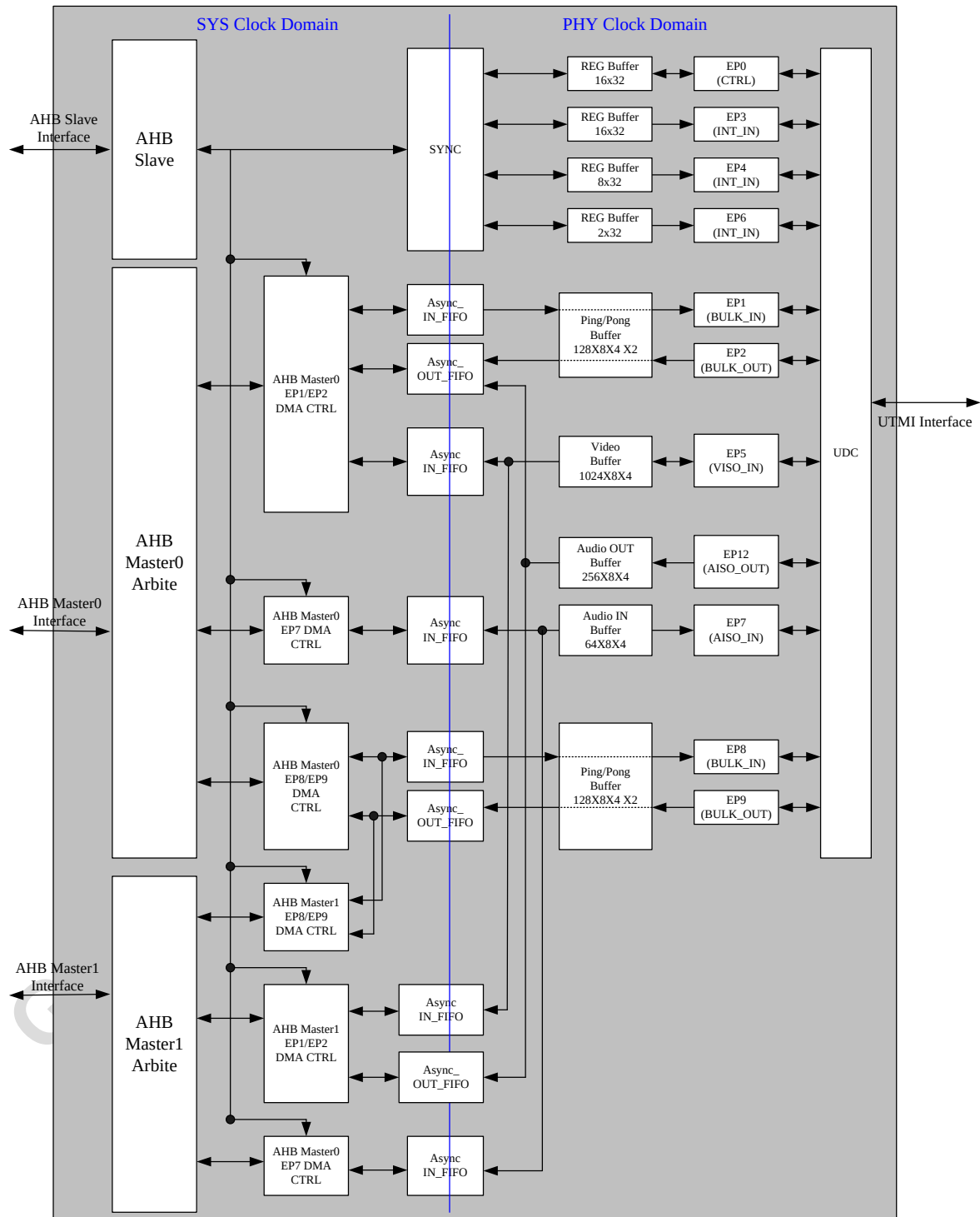


Figure 1.1 USB Device Block Diagram

1.4. Table of Interface/Alternate Setting for All Endpoints

The Endpoints in the device have their corresponding interface number and alternative setting, and according to the number and setting, there're also corresponding functions and limitations. For example, Endpoint 5 and Endpoint 7 are isochronous Endpoints, and there're many kinds of alternative setting (please refer to the following table) which decide the Max Package Size. User is able to configure the Endpoint interface according to the application or the parameters of Default. For interface 1~6, please refer to UDC_INTERFACE_MANUAL description of 0xD1100028/0xD2800028 and UDLC_INTERFACE description of 0xD11003C8/0xD28003C8. For interface 7 and Alternate, please refer to UDC_EP89_SET description of 0xD110052C/0xD280052C.

The following table shows the Default Endpoint, Type, Max Packet Size, Interface Number and Alternate Setting.

Table1.1 Interface and Alternate Map

EP Number	Endpoint Type	Max Packet Size	Interface Number	Alternate Setting
0	Control	64	0	0
1	BULK IN	512 (high speed) 64 (full speed)	Interface1[3:0] Default 0	0
2	BULK OUT	512 (high speed) 64 (full speed)	Interface1[3:0] Default 0	0
8	BULK IN	512 (high speed) 64 (full speed)	Interfac7[3:0] Default 0	Alternate7[3:0] Default 0
9	BULK OUT	512 (high speed) 64 (full speed)	Interfac7[3:0] Default 0	Alternate7[3:0] Default 0
3	INTERRUPT IN	64	Interface1[3:0] Default 0	0
4	INTERRUPT IN	32	Interface2[3:0] Default 1	0
6	INTERRUPT IN	8	Interface4[3:0] Default 3	0
c	Isochronous OUT	1024	Interfac6[3:0] Default 5	1
5	Isochronous IN	0	Interface3[3:0] Default 2	0
		256		1
		512		2
		1024 (high speed) 1020 (full speed)		3
		768*2(high speed)		4
		1024*2(high speed)		5
		852*3(high speed)		6
		1024*3(high speed)		7

EP Number	Endpoint Type	Max Packet Size	Interface Number	Alternate Setting
7	Isochronous IN	0	Interface5[3:0] Default 4	0
		16		1
		32		2
		64		3
		128		4
		256		5

1.5. Register Summary

Name	Address	Description
EP12DMACS_MAS0	0xD1100000/0xD2800000	Endpoint1 and Endpoint2 DMA Control Status for Master 0
EP12DMASADDR_MAS0	0xD1100004/0xD2800004	Endpoint1 and Endpoint2 DMA Start Address for Master 0
EP7DMACS_MAS0	0xD1100008/0xD2800008	Endpoint 7 DMA Control Status for Master 0
EP7DMASADDR_MAS0	0xD110000C/0xD280000C	Endpoint 7 DMA Start Address for Master 0
EP89DMACS_MAS0	0xD1100010/0xD2800010	Endpoint 8 and Endpoint 9 DMA Control Status for Master 0
EP89DMASADDR_MAS0	0xD1100014/0xD2800014	Endpoint 8 and Endpoint 9 DMA Start Address for Master 0
EP1011DMACS_MAS0	0xD1100018/0xD2800018	Endpoint 10 and Endpoint 11 DMA Control Status for Master 0
EP1011DMASADDR_MAS0	0xD110001C/0xD280001C	Endpoint 10 and Endpoint 11 DMA Start Address for Master 0
UDPHYI2CCR	0xD1100020/0xD2800020	PHY I2C Control
UDPHYI2CDR	0xD1100024/0xD2800024	PHY I2C Data
UDC_INTERFACE_MANUAL	0xD1100028/0xD2800028	UDC Interface Manual Mode Setting
EP12DMACS_MAS1	0xD1100040/0xD2800040	Endpoint1 and Endpoint2 DMA Control Status for Master 0
EP12DMASADDR_MAS1	0xD1100044/0xD2800044	Endpoint1 and Endpoint2 DMA Start Address for Master 1
EP7DMACS_MAS1	0xD1100048/0xD2800048	Endpoint 7 DMA Control Status for Master 1
EP7DMASADDR_MAS1	0xD110004C/0xD280004C	Endpoint 7 DMA Start Data Address for Master 1
EP89DMACS_MAS1	0xD1100050/0xD2800050	Endpoint 8 and Endpoint 9 DMA Control Status for Master 1
EP89DMASADDR_MAS1	0xD1100054/0xD2800054	Endpoint 8 and Endpoint 9 DMA Start Address for Master 1
EP1011DMACS_MAS1	0xD1100058/0xD2800058	Endpoint 10 and Endpoint 11 DMA Control Status for Master 1
EP1011DMASADDR_MAS1	0xD110005C/0xD280005C	Endpoint 10 and Endpoint 11 DMA Start Address for Master 1

Name	Address	Description
AHBMAS_SW	0xD1100060/0xD2800060	Master Switch
UPCS	0xD1100080/0xD2800080	USB PHY Control Status
UDCIE_MAS0	0xD1100084/0xD2800084	USB Device Controller Interrupt Enable for Master 0
UDCIF_MAS0	0xD1100088/0xD2800088	USB Device Controller Interrupt Flag for Master 0
UDC_IE_MAS1	0xD11000A0/0xD28000A0	USB Device Controller Interrupt Enable for Master 1
UDC_IF_MAS1	0xD11000A4/0xD28000A4	USB Device Controller Interrupt Flag for Master 1
EP4CTRL	0xD1100100/0xD2800100	Endpoint 4 Control
EP4PTR	0xD1100104/0xD2800104	Endpoint 4 FIFO Read write Pointer
EP4DATA	0xD1100108/0xD2800108	Endpoint 4 FIFO Data Port
EP4VB	0xD110010C/0xD280010C	Endpoint 4 Data Port Valid Byte
EP5HDEN	0xD1100140/0xD2800140	Endpoint 5 Header Data Enable
EP5HDLEN	0xD1100144/0xD2800144	Endpoint 5 Video Class Header Data Length
EP5FRAME	0xD1100148/0xD2800148	Endpoint 5 Video Class Frame Control
EP5HDCTRL	0xD110014C/0xD280014C	Endpoint 5 Video Class Header Data Control
EP5EN	0xD1100150/0xD2800150	Endpoint 5 Enable
EP5RPT	0xD1100154/0xD2800154	Endpoint 5 FIFO Read Pointer
EP5WPT	0xD1100158/0xD2800158	Endpoint 5 FIFO Write Pointer
EP5DATA	0xD110015C/0xD280015C	Endpoint 5 FIFO Data Port
EP5DMA_INTFCTRL	0xD1100160/0xD2800160	Endpoint 5 DMA and Interface Control
PE5SRE	0xD1100164/0xD2800164	USB Device SOF Recovery Enable
EP5FTLB	0xD1100168/0xD2800168	USB Device Frame Time Low Byte
EP5FTHB	0xD110016C/0xD280016C	USB Device Frame Time High Byte
EP5VB	0xD1100170/0xD2800170	Endpoint 5 Data Port Valid Byte
UDCINTF_ALT	0xD1100174/0xD2800174	USB Device Alternative Interface Setting
EP6CTRL	0xD1100180/0xD2800180	Endpoint 6 Control
EP6PTR	0xD1100184/0xD2800184	Endpoint 6 FIFO Write Pointer
EP6DATA	0xD1100188/0xD2800188	Endpoint 6 FIFO Data Port
EP6VB	0xD110018C/0xD280018C	Endpoint 6 Data Port Valid Byte
EP7CTRL	0xD11001C0/0xD28001C0	Endpoint 7 Control
EP7PTRR	0xD11001C4/0xD28001C4	Endpoint 7 FIFO Read Pointer
EP7PTR	0xD11001C8/0xD28001C8	Endpoint 7 FIFO Write Pointer
EP7DATA	0xD11001CC/0xD28001CC	Endpoint 7 FIFO Data Port
EP7VB	0xD11001D0/0xD28001D0	Endpoint 7 Data Port Valid Byte
EP5STCDIVL	0xD1100200/0xD2800200	Endpoint 5 Video Class Reference Clock Divisor Low Byte
EP5STCDIVH	0xD1100204/0xD2800204	Endpoint 5 Video Class Reference Clock Divisor High Byte
EP12C	0xD1100330/0xD2800330	Endpoint 1 and Endpoint 2 Control

Name	Address	Description
EP12PPC	0xD1100334/0xD2800334	Endpoint 1 and Endpoint 2 Ping-Pong FIFO Control
EP0VB	0xD1100340/0xD2800340	Endpoint 0 Data Port Valid Byte
EP0SETUPCS	0xD1100344/0xD2800344	Endpoint 0 Setup Control Status
EP0SETUPDP	0xD1100348/0xD2800348	Endpoint 0 Setup Data Port
EP0CS	0xD110034C/0xD280034C	Endpoint 0 Control Status
EP0DC	0xD1100350/0xD2800350	Endpoint 0 Data Count
EP0DP	0xD1100354/0xD2800354	Endpoint 0 Data Port
EP12PICL	0xD1100368/0xD2800368	Endpoint 1 and Endpoint 2 PING FIFO Data Count Low Byte
EP12PICH	0xD110036C/0xD280036C	Endpoint 1 and Endpoint 2 PING FIFO Data Count High Byte
EP12FDP	0xD1100370/0xD2800370	Endpoint 1 and Endpoint 2 FIFO Data Port
EP3CTRL	0xD1100374/0xD2800374	Endpoint 3 Control
EP3PTR	0xD1100378/0xD2800378	Endpoint 3 FIFO Write Pointer
EP3DATA	0xD110037C/0xD280037C	Endpoint 3 FIFO Data Port
EP3VB	0xD1100380/0xD2800380	Endpoint 3 Data Port Valid Byte
EPOONAKCN	0xD1100384/0xD2800384	Endpoint 0 OUT NAK Count
EPOINAKCN	0xD1100388/0xD2800388	Endpoint 0 IN NAK Count
EP1INAKCN	0xD110038C/0xD280038C	Endpoint 1 IN NAK Count
EP2ONAKCN	0xD1100390/0xD2800390	Endpoint 2 OUT NAK Count
EP12VB	0xD1100398/0xD2800398	Endpoint 1 and Endpoint 2 Data Port Valid Byte
EP12POCL	0xD110039C/0xD280039C	Endpoint 1 and Endpoint 2 PONG FIFO Data Count Low Byte
EP12POCH	0xD11003A0/0xD28003A0	Endpoint 1 and Endpoint 2 PONG FIFO Data Count High Byte
UDLCSET0	0xD11003B0/0xD28003B0	USB Device Linker Layer Controller Setting 0
UDLCSET1	0xD11003B4/0xD28003B4	USB Device Linker Layer Controller Setting 1
UDLCS	0xD11003B8/0xD28003B8	USB Device Linker Layer Controller Status
UDLCSTL	0xD11003BC/0xD28003BC	USB Device Linker Layer Controller Stall Control
UDLCSET2	0xD11003C0/0xD28003C0	USB Device Linker Layer Controller Setting 2
UDLCS2	0xD11003C4/0xD28003C4	USB Device Linker Layer Controller Status 2
UDLC_INTERFACE	0xD11003C8/0xD28003C8	USB Device Linker Layer Controller Interface
UDLCADDR	0xD11003F4/0xD28003F4	Device Address Register
UDC_IF	0xD1100400/0xD2800400	USB Device Linker Layer Interrupt Flag
UDC_IE	0xD1100404/0xD2800404	USB Device Linker Layer Interrupt Enable
USB_STD_IF	0xD1100410/0xD2800410	USB Standard Request Interrupt Flag
USB_STD_IE	0xD1100414/0xD2800414	USB Standard Request Interrupt Enable
UDC_NEW_IF	0xD1100420/0xD2800420	USB Device New Endpoint Interrupt Flag

Name	Address	Description
UDC_NEW_IE	0xD1100424/0xD2800424	USB Device New Endpoint Interrupt Enable
EP89C	0xD1100500/0xD2800500	Endpoint 8 and Endpoint 9 Control
EP89PPC	0xD1100504/0xD2800504	Endpoint 8 and Endpoint 9 Ping-Pong Control
EP89FS	0xD1100508/0xD2800508	Endpoint 8 and Endpoint 9 Status
EP89PICL	0xD110050C/0xD280050C	Endpoint 8 and Endpoint 9 Ping Count Low Byte
EP89PICH	0xD1100510/0xD2800510	Endpoint 8 and Endpoint 9 Ping Count High Byte
EP89POCL	0xD1100514/0xD2800514	Endpoint 8 and Endpoint 9 Pong Count Low Byte
EP89POCH	0xD1100518/0xD2800518	Endpoint 8 and Endpoint 9 Pong Count High Byte
EP89FDP	0xD110051C/0xD280051C	Endpoint 8 and Endpoint 9 Data Port
EP89VB	0xD1100520/0xD2800520	Endpoint 8 and Endpoint 9 Data Port Valid Byte
EP8INAKCN	0xD1100524/0xD2800524	Endpoint 8 IN NAK Count
EP9ONAKCN	0xD1100528/0xD2800528	Endpoint 9 OUT NAK Count
EP89_INTF	0xD110052C/0xD280052C	Endpoint 8 and Endpoint 9 Interface Setting
EPCER	0xD1100580/0xD2800580	Endpoint 12 Event
EPCCTRL	0xD1100584/0xD2800584	Endpoint 12 Control
EPCWCNT	0xD1100588/0xD2800588	Endpoint 12 Write Counter
EPCFR	0xD110058C/0xD280058C	Endpoint 12 Function
EPC_DATA	0xD1100590/0xD2800590	Endpoint 12 FIFO Data Port
IP_NAME	0xD1100FF8/0xD2800FF8	USB Device IP Name, ASCII: 5332_3232
VERSION	0xD1100FFC/0xD2800FFC	USB Device IP Version (2012-1214)

1.6. Register Definition

EP12DMACS_MAS0 **0xD1100000/0xD2800000** **Endpoint1 and Endpoint2 DMA Control Status for Master 0**

Bit	31	30	29	28	27	26	25	24
Function	DMA_EN	DMA_Flush_End	DMA_FIFO_Flush	DMA_Write	Modify_en	Count_alig_n	-	
Default	0	0	0	0	1	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-		DMA_ByteCount[21:16]					
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	DMA_ByteCount[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_ByteCount[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31]	DMA_EN	R/W	DMA Enable bit of Endpoint 1 or Endpoint 2. This bit will be cleared as 0 after DMA finishes all the process.	0: Idle. 1: Enable DMA of Endpoint 1 or Endpoint 2.
[30]	DMA_Flush_End	R/W	When DMA_FIFO_Flush = 1, DMA Flush will be activated. When DMA Flush finishes, this bit will be set as 1.	Read/Write 0: Don't care. Read 1: Endpoint 1 or Endpoint 2 flushing DMA FIFO finishes. Write 1: Clear this bit.
[29]	DMA_FIFO_Flush	R/W	Flush DMA FIFO of Endpoint 1 or Endpoint 2 When user writes 1 to this bit, it will reset Endpoint 1 or Endpoint 2 DMA function. This bit will be cleared as 0 after the Flush finishes.	0: Disable flushing DMA FIFO. 1: Enable flushing DMA FIFO.
[28]	DMA_Write	R/W	Selection of Endpoint 1 or Endpoint 2 Performing DMA Read or Write Function	0: DMA reads data from memory/IO. 1: DMA writes data to memory/IO.
[27]	Modify_en	R/W	For internal user only (function modify).	Always set this bit as 1.
[26]	Count_align	R/W	Endpoint 1 or Endpoint 2 Bulk DMA Length Alignment	0: 32 Bytes Alignment 1: HS, 512 bytes alignment. FS, 64 bytes alignment.
[25:22]	-	-	Reserved	-
[21:0]	DMA_ByteCount [21:0]	R/W	Endpoint 1 or Endpoint 2 DMA Transfer Length User can only write DMA_ByteCount when DMA_EN = 0, and DMA_ByteCount has to be the multiply of 32.	

EP12DMASADDR_MAS0 0xD1100004/0xD2800004 Endpoint1 and Endpoint2 DMA Start Address for Master 0

Bit	31	30	29	28	27	26	25	24
Function	DMA_SADDR[31:24]							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	DMA_SADDR[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	DMA_SADDR[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_SADDR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:0]	DMA_SADDR [31:0]	R/W	DMA Start address for Endpoint 1/Endpoint 2.	

EP7DMACS_MAS0 0xD1100008/0xD2800008 Endpoint7 DMA Control Status

Bit	31	30	29	28	27	26	25	24
Function	DMA_EN	DMA_Flush	-					
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	DMA_ByteCount[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_ByteCount[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31]	DMA_EN	R/W	DMA Enable bit of Endpoint 7 This bit will be cleared as 0 after DMA finishes all the process.	0: Idle. 1: Enable DMA of Endpoint 7.
[30]	DMA_Flush	R/W	Flush DMA FIFO of Endpoint 7 When user writes 1 to this bit, it will reset endpoint 7 DMA function. This bit will be cleared as 0 after the Flush finishes.	0: Disable flushing DMA FIFO. 1: Enable flushing DMA FIFO.

Bit	Function	Type	Description	Condition
[29:16]	-	-	Reserved	-
[15:0]	DMA_ByteCount [15:0]	R/W	Endpoint 7 DMA Transfer Length User can only write DMA_ByteCount when DMA_EN = 0, and DMA_ByteCount has to be the multiply of 32.	

EP7DMASADDR_MAS0 0xD110000C/0xD280000C Endpoint 7 DMA Start Address for Master 0

Bit	31	30	29	28	27	26	25	24
Function	DMA_SADDR[31:24]							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	DMA_SADDR[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	DMA_SADDR[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_SADDR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:0]	DMA_SADDR [31:0]	R/W	Endpoint 7 DMA Start Address	

**EP89DMACS_MAS0 0xD1100010/0xD2800010 Endpoint 8 and Endpoint 9 DMA Control Status
for Master 0**

Bit	31	30	29	28	27	26	25	24
Function	DMA_EN	DMA_Flush _End	DMA_FIFO _Flush	DMA_Write	Modify_en	Count_align	-	
Default	0	0	0	0	1	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-		DMA_ByteCount[21:16]					
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	DMA_ByteCount[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_ByteCount[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31]	DMA_EN	R/W	DMA Enable bit of Endpoint 8 or Endpoint 9 This bit will be cleared as 0 after DMA finishes all the process.	0: Idle. 1: Enable DMA of endpoint 8 or endpoint 9.
[30]	DMA_Flush_End	R/W	When DMA_FIFO_Flush = 1, DMA Flush will be activated. When DMA Flush finishes, this bit will be set as 1.	Read/Write 0: Don't care. Read 1: Endpoint 8 or Endpoint 9 flushing DMA FIFO finishes. Write 1: Clear this bit.
[29]	DMA_FIFO_Flush	R/W	Flush DMA FIFO of Endpoint 1 or Endpoint 2 When user writes 1 to this bit, it will reset endpoint 1 or endpoint 2 DMA function. This bit will be cleared as 0 after the Flush finishes.	0: Disable flushing DMA FIFO. 1: Enable flushing DMA FIFO.
[28]	DMA_Write	R/W	Selection of Endpoint 8 or Endpoint 9 Performing DMA Read or Write Function	0: DMA reads data from memory/IO. 1: DMA writes data to memory/IO.
[27]	Modify_en	R/W	For Device Bulk Function Configuration.	Always set this bit as 1.
[26]	Count_align	R/W	Bulk Out DMA Length Alignment.	0: 32 Bytes Alignment 1: HS, 512 bytes alignment. FS, 64 bytes alignment.
[25:22]	-	-	Reserved	-
[21:0]	DMA_ByteCount [21:0]	R/W	Endpoint 8 or Endpoint 9 DMA Transfer Length User can only write DMA_ByteCount when DMA_EN = 0, and DMA_ByteCount has to be the multiply of 32.	

EP89DMASADDR_MAS0
0xD1100014/0xD2800014
Endpoint 8 and Endpoint 9 DMA Start
Address for Master 0

Bit	31	30	29	28	27	26	25	24
Function	DMA_SADDR[31:24]							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	DMA_SADDR[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	DMA_SADDR[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_SADDR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:0]	DMA_SADDR [31:0]	R/W	DMA Start Address	

EP1011DMACS_MAS0
0xD1100018/0xD2800018
Endpoint 10 and Endpoint 11 DMA Control
Status for Master 0

Bit	31	30	29	28	27	26	25	24
Function	DMA_EN	DMA_Flush _End	DMA_FIFO _Flush	DMA_Writ e	Modify_en	Count_alig n	-	
Default	0	0	0	0	1	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-		DMA_ByteCount[21:16]					
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	DMA_ByteCount[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_ByteCount[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31]	DMA_EN	R/W	DMA Enable bit of Endpoint 10 or Endpoint 11 This bit will be cleared as 0 after DMA finishes all the process.	0: Idle. 1: Enable DMA of endpoint 10 or endpoint 11.
[30]	DMA_Flush_End	R/W	When DMA_FIFO_Flush = 1, DMA Flush will be activated. When DMA Flush finishes, this bit will be set as 1.	Read/Write 0: Don't care. Read 1: Endpoint 10 or Endpoint 11 flushing DMA FIFO finishes. Write 1: Clear this bit.
[29]	DMA_FIFO_Flush	R/W	Flush DMA FIFO of Endpoint 10 or Endpoint 11 When user writes 1 to this bit, it will reset endpoint 1 or endpoint 2 DMA function. This bit will be cleared as 0 after the Flush finishes.	0: Disable flushing DMA FIFO. 1: Enable flushing DMA FIFO.
[28]	DMA_Write	R/W	DMA Transfer Direction of Endpoint 10 or Endpoint 11	0: DMA reads data from memory/IO. 1: DMA writes data to memory/IO.
[27]	Modify_en	R/W	For internal user only (function modify).	Always set this bit as 1.
[26]	Count_align	R/W	Endpoint 10 or Endpoint 11 DMA Length Alignment.	0: 32 Bytes Alignment. 1: HS, 512 bytes alignment. FS, 64 bytes alignment.
[25:22]	-	-	Reserved	-
[21:0]	DMA_ByteCount [21:0]	R/W	Endpoint 10 or Endpoint 11 DMA Transfer Length User can only write DMA_ByteCount when DMA_EN = 0, and DMA_ByteCount has to be the multiply of 32.	

EP1011DMASADDR_MA 0xD110001C/0xD280001C Endpoint 10 and Endpoint 11 DMA Start SO Address for Master 0

Bit	31	30	29	28	27	26	25	24
Function	DMA_SADDR[31:24]							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	DMA_SADDR[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	DMA_SADDR[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_SADDR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:0]	DMA_ADDR[31:0]	R/W	DMA Start Address	

UDPHY2CCR 0xD1100020/0xD2800020 PHY I2C Control Register

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	CLKDIV[1:0]		-					SlaveAddr[6]
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	SlaveAddr[5:0]						PHYI2CWRn	PHYI2CEn
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:16]	-	-	Reserved	-
[15:14]	CLKDIV [1:0]	R/W	For I2C Clock selection: 2'b00: I2C_CLK = SYSCLK/32 2'b01: I2C_CLK = SYSCLK/64 2'b10: I2C_CLK = SYSCLK/128 2'b11: I2C_CLK = SYSCLK/256	
[13:9]	-	-	Reserved	-
[8:2]	SlaveAddr[6:0]	R/W	I2C Slave Address	
[1]	PHYI2CWRn	R/W	I2C Read/Write Control	0: I2C Read Data 1: I2C Write Data
[0]	PHYI2CEn	R/W	I2C_Start	0: Idle 1: I2C will start transmitting/receiving I2C Command to PHY. It will be auto

Bit	Function	Type	Description	Condition
				cleared when I2C transition finishes.

UDPHYI2CDR 0xD1100024/0xD2800024 PHY I2C Data Register

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	PHYADDR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	PHYDATA[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:16]	-	-	Reserved	-
[15:8]	PHYADDR[7:0]	R/W	I2C Address	
[7:0]	PHYDATA[7:0]	R/W	When PHYI2CWRn = 1 (Write Mode), I2C will transmit data. When PHYI2CWRn = 0 (Read Mode), I2C will receive data.	

UDC_INTERFAE_MANUAL 0xD1100028/0xD2800028 UDC Interface Manual Mode Setting

Bit	31	30	29	28	27	26	25	24
Function	IF_Manual_En							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	Interface6_value[3:0]				Interface5_manual[3:0]			
Default	0	1	0	1	0	1	0	0

Bit	15	14	13	12	11	10	9	8
Function	Interface4_manual[3:0]				Interface3_manual[3:0]			
Default	0	0	1	1	0	0	1	0

Bit	7	6	5	4	3	2	1	0
Function	Interface2_manual[3:0]				Interface1_manual[3:0]			
Default	0	0	0	1	0	0	0	0

Bit	Function	Type	Description	Condition
[31]	IF_Manual_En	R/W	Interface Manual Enable When this bit is set as 1, the interface1~5 Endpoint1~11 use will equal Interface1~5_manual. Or the interface1~5 will be configured according to UDLC_INTERFACE [4:0] (0xD11003C8/ 0xD28003C8). For the interface of corresponding EP, please refer to Table 1.1.	0: Interface manual mode is disabled. 1: Interface manual mode is enabled.
[30:24]	-	-	Reserved	-
[23:20]	Interface6_value [3:0]	R/W	The interface of EP12 equal Interface6_value.	
[19:16]	Interface5_manual [3:0]	R/W	Interface5 manual value. If IF_Manual_En = 1, The interface of EP7 equal Interface5_manual. (isochronous in for audio)	IF_Manual_En = 1, Interface5_manual value is valid
[15:12]	Interface4_manual [3:0]	R/W	Interface4 manual value. If IF_Manual_En = 1, The interface of EP6 equal Interface4_manual. (INT IN)	IF_Manual_En = 1, Interface4_manual value is valid
[11:8]	Interface3_manual [3:0]	R/W	Interface3 manual value. If IF_Manual_En = 1, The interface of EP5 equal Interface3_manual. (Video IOS IN)	IF_Manual_En = 1, Interface3_manual value is valid
[7:4]	Interface2_manual [3:0]	R/W	Interface2 manual value. If IF_Manual_En = 1, The interface of EP4 equal Interface2_manual. (INT IN)	IF_Manual_En = 1, Interface2_manual value is valid
[3:0]	Interface1_manual [3:0]	R/W	Interface1 Manual Value If IF_Manual_En = 1, The interface of EP1 (Bulk IN), EP2 (Bulk Out), EP3 (INT_IN) equal Interface1_manual.	IF_Manual_En = 1, Interface1_manual value is valid

EP12DMACS_MAS1 0xD1100040/0xD2800040 Endpoint 1 and Endpoint 2 DMA Control Status for Master 1

Bit	31	30	29	28	27	26	25	24
Function	DMA_EN	DMA_Flush_End	DMA_FIFO_Flush	DMA_Write	Modify_en	Count_alig_n	-	
Default	0	0	0	0	1	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-		DMA_ByteCount[21:16]					
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	DMA_ByteCount[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_ByteCount[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31]	DMA_EN	R/W	DMA Enable Bit of Endpoint 1 or Endpoint 2 This bit will be cleared as 0 after DMA finishes all the process.	0: Idle. 1: Enable DMA of endpoint 1 or endpoint 2.
[30]	DMA_Flush_End	R/W	When DMA_FIFO_Flush = 1, DMA Flush will be activated. When DMA Flush finishes, this bit will be set as 1.	Read/Write 0: Don't care. Read 1: Endpoint 1 or Endpoint 2 flushing DMA FIFO finishes. Write 1: Clear this bit.
[29]	DMA_FIFO_Flush	R/W	Flush DMA FIFO of Endpoint 1 or Endpoint 2 When user writes 1 to this bit, it will reset Endpoint 1 or Endpoint 2 DMA function. This bit will be cleared as 0 after the Flush finishes.	0: Disable flushing DMA FIFO. 1: Enable flushing DMA FIFO.
[28]	DMA_Write	R/W	Selection of Endpoint 1 or Endpoint 2 Performing DMA Read or Write Function	0: DMA reads data from memory/IO. 1: DMA writes data to memory/IO.
[27]	Modify_en	R/W	For internal user only (function modify).	Always set this bit as 1.
[26]	Count_align	R/W	Endpoint 1 or Endpoint 2 Bulk DMA Length Alignment	0: 32 Bytes Alignment 1: HS, 512 bytes alignment. FS, 64 bytes alignment.
[25:22]	-	-	Reserved	-
[21:0]	DMA_ByteCount [21:0]	R/W	Endpoint 1 or Endpoint 2 DMA Transfer Length User can only write DMA_ByteCount when DMA_EN = 0, and DMA_ByteCount has to be the multiply of 32.	

EP12DMASADDR_MAS1 0xD1100044/0xD2800044 Endpoint 1 and Endpoint 2 DMA Start Address for Master 1

Bit	31	30	29	28	27	26	25	24
Function	DMA_SADDR[31:24]							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	DMA_SADDR[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	DMA_SADDR[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_SADDR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:0]	DMA_SADDR [31:0]	R/W	DMA Start Address	

EP7DMACS_MAS1 0xD1100048/0xD2800048 Endpoint 7 DMA Control Status for Master 1

Bit	31	30	29	28	27	26	25	24
Function	DMA_EN	DMA_Flush_End	-					
Default	0	0	0	0	1	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	DMA_ByteCount[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_ByteCount[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31]	DMA_EN	R/W	DMA Enable Bit of Endpoint 7 This bit will be cleared as 0 after DMA finishes all the process.	0: Idle. 1: Enable DMA of Endpoint 7.
[30]	DMA_Flush	R/W	Flush DMA FIFO of Endpoint 7 When user writes 1 to this bit, it will reset Endpoint 7 DMA function. This bit will be cleared as 0 after the Flush finishes.	0: Disable flushing DMA FIFO. 1: Enable flushing DMA FIFO.
[29:16]	-	-	Reserved	-
[15:0]	DMA_ByteCount [15:0]	R/W	Endpoint 7 DMA Transfer Length User can only write DMA_ByteCount when DMA_EN = 0, and DMA_ByteCount has to be the multiply of 32.	

EP7DMASADDR_MAS1 0xD110004C/0xD280004C EP7 DMA Start Address for Master 1

Bit	31	30	29	28	27	26	25	24
Function	DMA_SADDR[31:24]							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	DMA_SADDR[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	DMA_SADDR[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_SADDR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:0]	DMA_SADDR [31:0]	R/W	DMA Start Address	

EP89DMACS_MAS1

0xD1100050/0xD2800050

Endpoint 8 and Endpoint 9 DMA Control Status
for Master 1

Bit	31	30	29	28	27	26	25	24
Function	DMA_EN	DMA_Flush_End	DMA_FIFO_Flush	DMA_Write	Modify_en	Count_align	-	
Default	0	0	0	0	1	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-		DMA_ByteCount[21:16]					
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	DMA_ByteCount[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_ByteCount[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31]	DMA_EN	R/W	DMA Enable Bit of Endpoint 8 or Endpoint 9 This bit will be cleared as 0 after DMA finishes all the process.	0: Idle. 1: Enable DMA of endpoint 8 or endpoint 9.
[30]	DMA_Flush_End	R/W	When DMA_FIFO_Flush = 1, DMA Flush will be activated. When DMA Flush finishes, this bit will be set as 1.	Read/Write 0: Don't care. Read 1: Endpoint 8 or Endpoint 9 flushing DMA FIFO finishes. Write 1: Clear this bit.
[29]	DMA_FIFO_Flush	R/W	Flush DMA FIFO of Endpoint 8 or Endpoint 9 When user writes 1 to this bit, it will reset Endpoint 8 or Endpoint 9 DMA function. This bit will be cleared as 0 after the Flush finishes.	0: Disable flushing DMA FIFO. 1: Enable flushing DMA FIFO.
[28]	DMA_Write	R/W	Selection of Endpoint 8 or Endpoint 9 Performing DMA Read or Write Function	0: DMA reads data from memory/IO. 1: DMA writes data to memory/IO.
[27]	Modify_en	R/W	For internal user only (function modify).	Always set this bit as 1.
[26]	Count_align	R/W	Endpoint 8 or Endpoint 9 Bulk DMA Length Alignment	0: 32 Bytes Alignment 1: HS, 512 bytes alignment.

Bit	Function	Type	Description	Condition
				FS, 64 bytes alignment.
[25:22]	-	-	Reserved	-
[21:0]	DMA_ByteCount [21:0]	R/W	Endpoint 8 or Endpoint 9 DMA Transfer Length User can only write DMA_ByteCount when DMA_EN = 0, and DMA_ByteCount has to be the multiply of 32.	

EP89DMASADDR_MAS1

0xD1100054/0xD2800054

Endpoint 8 and Endpoint 9 DMA Start Address

for Master 1

Bit	31	30	29	28	27	26	25	24
Function	DMA_SADDR[31:24]							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	DMA_SADDR[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	DMA_SADDR[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_SADDR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:0]	DMA_ADDR[31:0]	R/W	DMA Start Address	

EP1011DMACS_MAS1

0xD1100058/0xD2800058

Endpoint 10 and Endpoint 11 DMA Control

Status for Master 1

Bit	31	30	29	28	27	26	25	24
Function	DMA_EN	DMA_Flush _End	DMA_FIFO _Flush	DMA_Writ e	Modify_en	Count_alig n	-	
Default	0	0	0	0	1	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-		DMA_ByteCount[21:16]					
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	DMA_ByteCount[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_ByteCount[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31]	DMA_EN	R/W	DMA Enable Bit of Endpoint 10 or Endpoint 11 This bit will be cleared as 0 after DMA finishes all the process.	0: Idle. 1: Enable DMA of endpoint 10 or endpoint 11.
[30]	DMA_Flush_End	R/W	When DMA_FIFO_Flush = 1, DMA Flush will be activated. When DMA Flush finishes, this bit will be set as 1.	Read/Write 0: Don't care. Read 1: Endpoint 10 or Endpoint 11 flushing DMA FIFO finishes. Write 1: Clear this bit.
[29]	DMA_FIFO_Flush	R/W	DMA FIFO of Flush endpoint 10 or Endpoint 11 When user writes 1 to this bit, it will reset Endpoint 10 or Endpoint 11 DMA function. This bit will be cleared as 0 after the Flush finishes.	0: Disable flushing DMA FIFO. 1: Flush DMA FIFO.
[28]	DMA_Write	R/W	Selection of Endpoint 10 or Endpoint 11 Performing DMA Read or Write Function	0: DMA reads data from memory/IO. 1: DMA writes data to memory/IO.
[27]	Modify_en	R/W	For internal user only (function modify).	Always set this bit as 1.
[26]	Count_align	R/W	Endpoint 10 or Endpoint 11 Bulk DMA Length Alignment	0: 32 Bytes Alignment 1: HS, 512 bytes alignment. FS, 64 bytes alignment.
[25:22]	-	-	Reserved	-
[21:0]	DMA_ByteCount [21:0]	R/W	Endpoint 10 or Endpoint 11 DMA Transfer Length User can only write DMA_ByteCount when DMA_EN = 0, and DMA_ByteCount has to be the multiply of 32.	

EP1011DMASADDR_MA 0xD110005C/0xD280005C
Endpoint 10 and Endpoint 11 DMA Start
S1
Address for Master 1

Bit	31	30	29	28	27	26	25	24
Function	DMA_SADDR[31:24]							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	DMA_SADDR[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	DMA_SADDR[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DMA_SADDR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:0]	DMA_ADDR [31:0]	R/W	DMA Address	

AHBMAS_SW
0xD1100060/0xD2800060
Master Switch

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-		EP12_MAS_SW	EP5_MAS_SW	EP7_MAS_SW	-	EP8/9_MAS_SW	-
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:6]	-	-	Reserved	-
[5]	EP12_MAS_SW	R/W	Endpoint 12 Master Switch	0: Endpoint12 uses Master 0. 1: Endpoint12 uses Master 1.
[4]	EP5_MAS_SW	R/W	Endpoint 5 Master Switch	1: Endpoint5 uses Master 1. 0: Endpoint5 uses Master 0.
[3]	EP7_MAS_SW	R/W	Endpoint 7 Master Switch	1: Endpoint7 uses Master 1. 0: Endpoint7 uses Master 0.
[2]	-	-	Reserved	-
[1]	EP8/9_MAS_SW	R/W	Endpoint 8 and Endpoint 9 Master Switch	1: Endpoint8/9 uses Master 1. 0: Endpoint8/9 uses Master 0.
[0]	-	-	Reserved	-

UPCS
0xD1100080/0xD2800080
USB PHY Control Status

Bit	31	30	29	28	27	26	25	24
Function	DISCONNECT	USB_CLK_RDY	PARTIAL	SUSPENDM	FORCE_DISCONNECT	FORCE_CONNECT	PHYCLK_SEL	PHYCLK_READY
Default	1	0	1	1	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	Sync_sel	-				SFT_USBRST	-	
Default	0	0	0	0	0	0	0	1

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31]	DISCONNECT	R/W	This bit is used to force USB PHY to disconnect from USB bus.	0: Connect, that is DP is pulled high and DM is pulled low. 1: Disconnect, that is, DP and DM are pulled low.
[30]	USB_CLK_RDY	R	USB Clock Ready Status Signal When PHYCLK_READY =1, USB_CLK_RDY will	0: USB Device PHY clock is not ready.

Bit	Function	Type	Description	Condition
			be set as high.	1: USB Device PHY clock is ready.
[29]	PARTIAL	R/W	USB PHY Partial Control Signal	0: USB PHY uses all the clocks. 1: USB PHY uses partial clocks.
[28]	SUSPENDM	R/W	When user sets this bit as 0, USB PHY will suspend.	0: Suspend. 1: Normal.
[27]	FORCE_DISCONNECT	R/W	Write 1 to this bit, and it will force the USB bus to disconnect with the host. It's not suggested to use this function.	0: Connect. 1: Disconnect.
[26]	FORCE_CONNECT	R/W	Write 1 to this bit, and it will force the USB bus to disconnect with the host. It's not suggested to use this function.	0: Disconnect. 1: Connect.
[25]	PHYCLK_SEL	R/W	This bit is used to select device PHY clock which is from slow clock or PHY Macro clock.	0: PHY Clock 1: Slow Clock
[24]	PHYCLK_READY	R	PHYCLK_READY signal is from PHY Macro. When this bit is 1, it means PHY clock is ready.	0: PHY clock is not ready. 1: PHY clock is ready.
[23]	Sync_sel	R/W	UDC Read/Write Points Sync Clock Selection When sync_sel = 1, UDC Read/write points sync clock needs 2T. When sync_sel = 0, UDC Read/write points sync clock needs 1T. It's suggested to fix this bit as 0.	0: 1T 1: 2T
[22:19]	-	-	Reserved	-
[18]	SFT_USBRST	R/W	USB Device Software Reset Write 1 to this bit will execute Software reset. SFT_USBRST will be cleared automatically when the reset finishes.	0: Software Reset finishes. 1: Execute Software Reset.
[17:0]	-	-	Reserved	-

UDCIE_MAS0
0xD1100084/0xD2800084
USB Device Controller Interrupt Enable for Master 0

Bit	31	30	29	28	27	26	25	24
Function	-	EPAB_DMA_IE	EP89_DMA_IE	AUDIO_DMA_IE	FORCE_DISCONNECT_IE	FORCE_CONNECT_IE	EP12_DMA_IE	VBUS_IE
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	UDC_IE[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	UDC_IE[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	UDC_IE[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31]	-	-	Reserved	-
[30]	EPAB_DMA_IE	R/W	Endpoint 10 or Endpoint 11 DMA Finish Interrupt Enable	0: Disable 1: Enable
[29]	EP89_DMA_IE	R/W	Endpoint 8 or Endpoint 9 DMA Finish Interrupt Enable	0: Disable 1: Enable
[28]	AUDIO_DMA_IE	R/W	Audio (Endpoint 7) DMA Finish Interrupt Enable	0: Disable 1: Enable
[27]	FORCE_DISCONNECT_IE	R/W	Force USB Bus Disconnect Interrupt Enable	0: Disable 1: Enable
[26]	FORCE_CONNECT_IE	R/W	Force USB Bus Connect Interrupt Enable	0: Disable 1: Enable
[25]	EP12_DMA_IE	R/W	Endpoint 1, Endpoint 2, Endpoint 5 or Endpoint 12 DMA Finish Interrupt Enable	0: Disable 1: Enable
[24]	VBUS_IE	R/W	VBUS Interrupt Enable	0: Disable 1: Enable
[23:0]	UDC_IE [23:0]	R	USB Device Control Interrupt Enable The definition is the same as 0xD1100404/0xD2800404[23:0].	0: Disable 1: Enable

UDCIF_MAS0
0xD1100088/0xD2800088
USB Device Controller Interrupt Flag for
Master 0

Bit	31	30	29	28	27	26	25	24
Function	-	EPAB_DMA_IF	EP89_DMA_IF	AUDIO_DMA_IF	FORCE_DISCONNECT_IF	FORCE_CONNECT_IF	EP12_DMA_IF	VBUS_IF
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	UDC_IF[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	UDC_IF[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	UDC_IF[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31]	-	-	Reserved	-
[30]	EPAB_DMA_IF	R/W	Endpoint 10 or Endpoint 11 DMA Finish Interrupt Flag Write 1 to clear this bit.	0: Interrupt flag is not asserted. 1: Interrupt flag is asserted.
[29]	EP89_DMA_IF	R/W	Endpoint 8 or Endpoint 9 DMA Finish Interrupt Flag If the interrupt flag is asserted, it means DMA finishes all the process. Generally, user has to check this bit to see whether the OUT (endpoint9) DMA transfer is done or not. Write 1 to clear this bit.	0: Interrupt flag is not asserted. 1: Interrupt flag is asserted.
[28]	AUDIO_DMA_IF	R/W	Audio (Endpoint 7) DMA Finish Interrupt Flag If the interrupt flag is asserted, it means DMA finishes all the process. Write 1 to clear this bit.	0: Interrupt flag is not asserted. 1: Interrupt flag is asserted.
[27]	FORCE_DISCONNECT_IF	R/W	Force USB Bus to Disconnect Interrupt Flag Write 1 to clear this bit.	0: Interrupt flag is not asserted. 1: Interrupt flag is asserted.
[26]	FORCE_CONNECT_IF	R/W	Force USB Bus Connect Interrupt Flag Write 1 to clear this bit.	0: Interrupt flag is not asserted. 1: Interrupt flag is asserted.
[25]	EP12_DMA_IF	R/W	Endpoint 1, Endpoint 2, Endpoint 5 or Endpoint 12 DMA Finish Interrupt Flag If the interrupt flag is asserted, it means DMA finishes all the process. Generally, user has to check this bit to see whether the OUT (endpoint2 and endpoint 5) DMA transfer is done or not. Write 1 to clear this bit.	0: Interrupt flag is not asserted. 1: Interrupt flag is asserted.
[24]	VBUS_IF	R/W	VBUS Interrupt Flag Write 1 to clear this bit.	0: Interrupt flag is not asserted. 1: Interrupt flag is asserted.
[23:0]	UDC_IF	-	USB Device Control Interrupt Flag	0: Interrupt flag is not

Bit	Function	Type	Description	Condition
	[23:0]		For description of each bit, please refer to 0xD1100400/0xD2800400[23:0]. Write 1 to clear this bit.	asserted. 1: Interrupt flag is asserted.

UDC_IE_MAS1

0xD11000A0/0xD28000A0

USB Device Controller Interrupt Enable for Master 1

Bit	31	30	29	28	27	26	25	24
Function	-	EPAB_DMA_IE_MAS1	EP89_DMA_IE_MAS1	EP7_DMA_IE_MAS1	-	-	EP12_DMA_IE_MAS1	-
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-	-	-	-	-	-	-	-
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-	-	-	-	-	-	-	-
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-	-	-	-	-	-	-	-
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31]	-	-	Reserved	-
[30]	EPAB_DMA_IEMAS1	R/W	Endpoint 10 or endpoint 11 DMA finish interrupt enable for master 1.	0: Disable 1: Enable
[29]	EP89_DMA_IEMAS1	R/W	Endpoint 8 or endpoint 9 DMA finish interrupt enable for master 1.	0: Disable 1: Enable
[28]	EP7_DMA_IEMAS1	R/W	Endpoint 7 DMA finish interrupt enable for master 1.	0: Disable 1: Enable
[27:26]	-	-	Reserved	-
[25]	EP12_DMA_IE_MAS1	R/W	Endpoint 1, endpoint 2, endpoint 5 or endpoint 12 DMA finish interrupt enable for master 1.	0: Disable 1: Enable
[24:0]	-	-	Reserved	-

UDC_IF_MAS1 0xD11000A4/0xD28000A4 USB Device Controller Interrupt Flag for Master 1

Bit	31	30	29	28	27	26	25	24
Function	-		EP89_DMA_IF_MAS1	Audio_DMA_IF_MAS1	-		EP12_DMA_IF_MAS1	-
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:30]	-	-	Reserved	-
[29]	EP89_DMA_IF_MAS1	R/W	Endpoint 8 and Endpoint 9 DMA finish interrupt flag for master 1. It means even DMA finishes all the process, for USB IN data, USB controller has to wait for the ACK from USB host, and the transfer is done. Write 1 to clear this bit.	0: Interrupt flag is not asserted. 1: Interrupt flag is asserted.
[28]	Audio_DMA_IF_MAS1	R/W	Audio (Endpoint 7) DMA finish interrupt flag for master 1. It means even DMA finishes all the process, for USB IN data, USB controller has to wait for the ACK from USB host, and the transfer is done. Write 1 to clear this bit.	0: Interrupt flag is not asserted. 1: Interrupt flag is asserted.
[27:26]	-	-	Reserved	-
[25]	EP12_DMA_IF_MAS1	R/W	Endpoint 1, Endpoint 2, Endpoint 5 or Endpoint 12 DMA finish interrupt flag for master 1. It means even DMA finishes all the process, for USB IN data, USB controller has to wait for the ACK from USB host, and the transfer is done. Write 1 to clear this bit.	0: Interrupt flag is not asserted. 1: Interrupt flag is asserted.
[24:0]	-	-	Reserved	-

EP4CTRL	0xD1100100/0xD2800100						Endpoint 4 Control	
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EP4_INTF[3:0]				EP4_VLD	-		EP4_EN
Default	0	0	0	1	0	0	0	1

Bit	Function	Type	Description	Condition
[31:8]	-	-	Reserved	-
[7:4]	EP4_INTF [3:0]	R/W	Endpoint 4 Interface Number This value should be the same as 0xD1100028 /0xD2800028 [7:4].	
[3]	EP4_VLD	R/W	Endpoint 4 Data Valid When Endpoint 4 IN data is ready (valid), user has to set this bit as 1 to trigger USB device controller to wait for the IN data of the host. When USB device controller receives the ACK from the host, this bit will be cleared automatically.	0: The data is not valid. 1: The data is valid.
[2:1]	-	-	Reserved	-
[0]	EP4_EN	R/W	Endpoint 4 Enable Bit	0: Disable Endpoint 4. 1: Enable Endpoint 4.

EP4PTR	0xD1100104/0xD2800104						Endpoint 4 FIFO Read Write Pointer	
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	UDEP4PTR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:8]	-	-	Reserved	-
[7:0]	UDEP4PTR [7:0]	R/W	USB Device Endpoint 4 FIFO Read and Write Pointer	

EP4DATA 0xD1100108/0xD2800108 Endpoint 4 FIFO Data Port

Bit	31	30	29	28	27	26	25	24
Function	UDEP4DATA[31:24]							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	UDEP4DATA[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	UDEP4DATA[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	UDEP4DATA[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:0]	UDEP4DATA [31:0]	R/W	USB Device Endpoint 4 FIFO Data Port For the valid Bytes of reading or writing UDEP4DATA,	

Bit	Function	Type	Description	Condition
			please refer to EP4VB (0xD110010C /0xD2800010C).	

EP4VB 0xD110010C/0xD280010C Endpoint 4 Data Port Valid Byte

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-				UDEP4VB[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:4]	-	-	Reserved	-
[3:0]	UDEP4VB [3:0]	R/W	USB Device Endpoint 4 Data Port Valid Byte UDEP4VB indicates which Byte(s) is valid in UDEP4DATA (0xD1100108 /0xD28000108). Data Port [31:0] = {Byte3, Byte2, Byte1, Byte0}	UDEP4VB = 0: No byte is valid. UDEP4VB[0] = 1: Byte0 is valid. UDEP4VB[1] = 1: Byte1 is valid. UDEP4VB[2] = 1: Byte2 is valid. UDEP4VB[3] = 1: Byte3 is valid.

EP5HDEN 0xD1100140/0xD2800140 Endpoint 5 Header Data Enable

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-			NullData_Nohead	-		UDEP5FLUSH	UDEP5HDEN
Default	0	0	0	0	0	0	0	1

Bit	Function	Type	Description	Condition
[31:5]	-	-	Reserved	-
[4]	NullData_Nohead	R/W	When Endpoint5 is sending NullData IN, it will not send header Data to the host. This bit is valid only when UDEP5HDEN = 1.	0: NullData_Nohead is disabled. 1: NullData_Nohead is enabled.
[3:2]	-	-	Reserved	-
[1]	UDEP5FLUSH	R/W	USB Device Endpoint 5 FIFO Flush Bit This bit will not be cleared automatically. User has to write 0 to this bit manually after flush.	0: Disable endpoint 5 flush. 1: Enable endpoint 5 flush.
[0]	UDEP5HDEN	R/W	EP5 Header Enable When this bit set as 1, USB device will send UVC Header Data automatically.	0: Disable 1: Enable

EP5HDLEN 0xD1100144/0xD2800144 Endpoint 5 Video Class Header Data Length

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	UDEP5HDLEN[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition										
[31:8]	-	-	Reserved											
[7:0]	UDEP5HDLEN [7:0]	R/W	Length of UVC Payload Header (Byte) This value is usually 12. Please refer to Universal Serial Bus Device Class Definition for Video Devices. For the configuration of UDEP5HDLEN, it depends on EP5HDFMT[1:0] (0xD110014C[1:0] / 0xD280014C[1:0]). <table><thead><tr><th>EP5HDFMT[1:0]</th><th>UDEP5HDLEN (Byte)</th></tr></thead><tbody><tr><td>00</td><td>2</td></tr><tr><td>01</td><td>6</td></tr><tr><td>10</td><td>8</td></tr><tr><td>11</td><td>12</td></tr></tbody></table>	EP5HDFMT[1:0]	UDEP5HDLEN (Byte)	00	2	01	6	10	8	11	12	
EP5HDFMT[1:0]	UDEP5HDLEN (Byte)													
00	2													
01	6													
10	8													
11	12													

EP5FRAME 0xD1100148/0xD2800148 Endpoint 5 Video Class Frame Control

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EP5CTRL EMPTY	EP5CTRLFU LL	-			UDEP5END FR_Auto	UDEP5END FR	UDEP5FRA MEID
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:8]	-	-	Reserved	-
[7]	EP5CTRLEMPTY	R	Endpoint 5 Buffer Empty	0: Endpoint 5 buffer is not empty. 1: Endpoint 5 buffer is empty.
[6]	EP5CTRLFULL	R	Endpoint 5 Buffer Full	0: Endpoint 5 buffer is not full. 1: Endpoint 5 buffer is full.

Bit	Function	Type	Description	Condition
[5:3]	-	-	Reserved	-
[2]	UDEP5ENDFR_Auto	R/W	USB controller will automatically generate the “End of Frame” bit of UVC payload header. Please refer to Universal Serial Bus Device Class Definition for Video: Chapter 2.4.3.3.	0: The bit will not be generated automatically, and user has to set UDEP5ENDFR as 1 manually. 1: The bit will be generated automatically.
[1]	UDEP5ENDFR	R/W	“End of Frame” bit of UVC Payload Header Configuration Please refer to Universal Serial Bus Device Class Definition for Video: Chapter 2.4.3.3.	0: “End of Frame” bit of UVC payload header is 0. 1: “End of Frame” bit of UVC payload header is 1.
[0]	UDEP5FRAMEID	R/W	Frame ID of UVC Payload Header Please refer to Universal Serial Bus Device Class Definition for Video Devices: Chapter 2.4.3.3.	0: Frame ID of UVC payload header is 0. Write 1 to toggle UVC frame ID. 1: Frame ID of UVC payload header is 1.

EP5HDCtrl 0xD110014C/0xD280014C Endpoint 5 Video Class Header Data Control

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-			UDEP5ErrFlg	UDEP5FRAMEND	UDEP5FRAMESTILL	UDEP5FRAMESCR	UDEP5FRAMEPT
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:5]	-	-	Reserved	-
[4]	UDEP5ErrFlg	R/W	Error Flag of UVC Payload Header Please refer to Universal Serial Bus Device Class	0: Error flag of UVC payload header is 0.

Bit	Function	Type	Description	Condition										
			Definition for Video Devices: Chapter 2.4.3.3.	1: Error flag of UVC payload header is 1.										
[3]	UDEP5FRAMEND	R/W	Header Data End Flag for Header Data If this bit is set as 1, the UVC payload header generated by the HW will set the End of header bit in bmHeaderInfo field as 1.											
[2]	UDEP5FRAMESTILL	R/W	Header Data Still Image Flag for Header Data If this bit is set as 1, the UVC payload header generated by the HW will set the Still Image bit in bmHeaderInfo field as 1.											
[1:0]	EP5HDFMT	R/W	EP5 Hardware Header Format EP5HDFMT determines the length and format of the header data. Header Data length: <table border="1"><thead><tr><th>EP5HDFMT[1:0]</th><th>UDEP5HDLEN (Byte)</th></tr></thead><tbody><tr><td>00</td><td>2</td></tr><tr><td>01</td><td>6</td></tr><tr><td>10</td><td>8</td></tr><tr><td>11</td><td>12</td></tr></tbody></table> For header format, please refer to Universal Serial Bus Device Class Definition for Video Devices.	EP5HDFMT[1:0]	UDEP5HDLEN (Byte)	00	2	01	6	10	8	11	12	
EP5HDFMT[1:0]	UDEP5HDLEN (Byte)													
00	2													
01	6													
10	8													
11	12													

Note:

- When the Frame ID of UVC payload header toggles, the hardware will write the values of stlkcnt[31:0] (4 bytes) to the PTS of UVC payload header.
- SCR (source time control register) [47:0] of UVC payload header = {5'b0, sof_cnt[10:0], stc_reg[31:0]}

EP5EN 0xD1100150/0xD2800150 Endpoint 5 Enable

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-			EP5_auto_gen _frame_id	EP5_Auto_ Mode	-		UDEP5EN
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:5]	-	-	Reserved	-
[4]	EP5_auto_gen_fra me_id	R/W	Endpoint5 frame ID Auto-Generation The hardware will generate frame ID automatically when a frame ends.	0: Disable 1: Enable
[3]	EP5_Auto_Mode	R/W	Endpoint 5 auto mode for JPEG function. This is a special function used when user wants to accelerate the JPEG data transfer from USB device to the host.	0: Disable Auto Mode. 1: Enable Auto Mode.
[2:1]	-	-	Reserved	-
[0]	UDEP5EN	R/W	USB Device Endpoint 5 Enable	0: Disable 1: Enable

EP5RPT
0xD1100154/0xD2800154
Endpoint 5 FIFO Read Pointer

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-			UDEP5RPT[12:8]				
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	UDEP5RPT[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:13]	-	-	Reserved	-
[12:0]	UDEP5RPT[12:0]	R/W	USB Device Endpoint 5 FIFO Read Pointer	

EP5WPT		0xD1100158/0xD2800158				Endpoint 5 FIFO Write Pointer		
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit		23	22	21	20	19	18	17	16
Function	-								
Default		0	0	0	0	0	0	0	0

Bit		15	14	13	12	11	10	9	8
Function		-			UDEP5WPT[12:8]				
Default		0	0	0	0	0	0	0	0

Bit		7	6	5	4	3	2	1	0
Function		UDEP5WPT[7:0]							
Default		0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:13]	-	-	Reserved	-
[12:0]	UDEP5WPT [12:0]	R/W	USB Device Endpoint 5 FIFO Write Pointer	

EP5DATA		0xD110015C/0xD280015C					Endpoint 5 FIFO Data Port	
Bit	31	30	29	28	27	26	25	24
Function	UDEP5DATA[31:24]							
Default	0	0	0	0	0	0	0	0

Bit		23	22	21	20	19	18	17	16
Function	UDEP5DATA[23:16]								
Default		0	0	0	0	0	0	0	0

Bit		15	14	13	12	11	10	9	8
Function	UDEP5DATA[15:8]								
Default		0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	UDEP5DATA[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:0]	UDEP5DATA [31:0]	R/W	USB Device Endpoint 5 FIFO Data Port For how many valid reading or writing byte of data is valid, please refer to EP5VB (0xD1100170 /0xD28000170).	

EP5DMA_INTFCTRL 0xD1100160/0xD2800160 Endpoint5 DMA and Interface Control

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	--							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	UDC_INTF_EP5[3:0]				ALTINT_CONFIG_REAL[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	ALTINT_CONFIG[3:0]				-			UDVDMAEN
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:16]	-	-	Reserved	-
[15:12]	UDC_INTF_EP5 [3:0]	R/W	EP5 Interface Number When the host issues a "Set_Interface" command to EP5, the interface of EP5 will be configured. Note: The interface mentioned above is Interface3 of EP5. For details, please refer to Table 1.1.	
[11:8]	ALTINT_CONFIG _REALL[3:0]	R	Real Alter Interface Information for USB Controller	
[7:4]	ALTINT_CONFIG _EP5[3:0]	R/W	EP5 Alter Interface Information The value of ALTINT_CONFIG_EP5 defines the maximum package size every SOF of Endpoint5 can transfer.	4'h0: The package size is 0. 4'h1: The maximum package size is 256 Byte.

Bit	Function	Type	Description	Condition
			When the host issues a "Set_Interface" command to EP5, the Alter interface of EP5 will be configured. Note: The interface mentioned above is Interface3 of EP5. For details, please refer to Table 1.1.	4'h2: The maximum package size is 512 Byte 4'h3: The maximum package size is 1024 Byte 4'h4: The maximum package size is 1536 Byte 4'h5: The maximum package size is 2048 Byte 4'h6: The maximum package size is 2556 Byte 4'h7: The maximum package size is 3072 Byte
[3:1]	-	-	Reserved	-
[0]	UDVDMAEN	R/W	Endpoint 5 DMA Enable	0: Disable 1: Enable

EP5SRE

0xD1100164/0xD2800164

USB Device SOF Recovery Enable

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-							UDSRE
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:1]	-	-	Reserved	-
[0]	UDSRE	R/W	USB Device SOF Recovery Enable In case the host SOF is not accurate, a SOF generated by the USB device will be provided to Endpoint5.	0: Bypass sof_org. 1: SOF recovery is enabled. (SCR)

EP5FTLB		0xD1100168/0xD2800168					USB Device Frame Time Low Byte	
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit		23	22	21	20	19	18	17	16
Function	-								
Default		0	0	0	0	0	0	0	0

Bit		15	14	13	12	11	10	9	8
Function	-								
Default		0	0	0	0	0	0	0	0

Bit		7	6	5	4	3	2	1	0
Function	UDFTLB[7:0]								
Default		1	0	1	0	0	1	1	0

Bit	Function	Type	Description	Condition
[31:8]	-	-	Reserved	
[7:0]	UDFTLB [7:0]	R/W	USB Device Frame Time Low Byte Default value for PHY clock is 30M, and device HS 125us/33/33ns = 0xEA6.	

EP5FTHB		0xD110016C/0xD280016C				USB Device Frame Time High Byte		
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit		23	22	21	20	19	18	17	16
Function	-								
Default		0	0	0	0	0	0	0	0

Bit		15	14	13	12	11	10	9	8
Function	-								
Default		0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	UDFTHB[7:0]							
Default	0	0	0	0	1	1	1	0

Bit	Function	Type	Description	Condition
[31:8]	-	-	Reserved	-
[7:0]	UDFTHB [7:0]	R/W	USB Device Frame Time High Byte Default value for PHY clock is 30M, and device HS 125us/33/33ns = 0xEA6.	

EP5VB
0xD1100170/0xD2800170
Endpoint 5 Data Port Valid Byte

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-				UDEP5VB[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:4]	-	-	Reserved	-
[3:0]	UDEP5VB [3:0]	R/W	USB Device Endpoint5 Data Port Valid Byte UDEP5VB indicates which Byte(s) is valid in UDEP5DATA (0xD110015C/0xD280015C). Data Port [31:0] = {Byte3, Byte2, Byte1, Byte0}.	UDEP5VB = 0: No byte is valid. UDEP5VB[0] = 1: Byte0 is valid. UDEP5VB[1] = 1: Byte1 is valid. UDEP5VB[2] = 1: Byte2 is valid. UDEP5VB[3] = 1: Byte3 is valid.

UDCINTF_ALT		0xD1100174/0xD2800174				USB Device Alternative Interface Setting		
Bit	31	30	29	28	27	26	25	24
Function	-				USBINTF6_ALT[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	USBINTF5_ALT[3:0]				USBINTF4_ALT[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	USBINTF3_ALT[3:0]				USBINTF2_ALT[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	USBINTF1_ALT[3:0]				USBINTF0_ALT[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:28]	-	-	Reserved	-
[27:24]	USBINTF6_ALT[3:0]	R	The alternative setting value for interface 6.	
[23:20]	USBINTF5_ALT[3:0]	R	The alternative setting value for interface 5.	
[19:16]	USBINTF4_ALT[3:0]	R	The alternative setting value for interface 4.	
[15:12]	USBINTF3_ALT[3:0]	R	The alternative setting value for interface 3.	
[11:8]	USBINTF2_ALT[3:0]	R	The alternative setting value for interface 2.	
[7:4]	USBINTF1_ALT[3:0]	R	The alternative setting value for interface 1.	
[3:0]	USBINTF0_ALT[3:0]	R	The alternative setting value for interface 0.	

EP6CTRL		0xD1100180/0xD2800180					Endpoint 6 Control	
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-				EP6_VLD	-		EP6_EN
Default	0	0	0	0	0	0	0	1

Bit	Function	Type	Description	Condition
[31:4]	-	-	Reserved	-
[3]	EP6_VLD	R/W	Endpoint 6 Data Valid When Endpoint 6 IN data is ready (valid), user has to set this bit as 1 to trigger USB Device controller to wait for the IN data of the host. When USB device controller receives the ACK from the host, this bit will be cleared automatically.	0: The data is not valid. 1: The data is valid.
[2:1]	-	-	Reserved	-
[0]	EP6_EN	R/W	Endpoint 6 Enable Bit	0: Disable endpoint 6. 1: Enable endpoint 6.

EP6PTR 0xD1100184/0xD2800184 Endpoint 6 FIFO Read Write Pointer

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	UDEP6PTR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:8]	-	-	Reserved	-
[7:0]	UDEP6PTR[7:0]	R/W	USB Device Endpoint 6 FIFO Read Write Pointer	

EP6DATA **0xD1100188/0xD2800188** **Endpoint 6 FIFO Data Port**

Bit	31	30	29	28	27	26	25	24
Function	UDEP6DATA[31:24]							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	UDEP6DATA[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	UDEP6DATA[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	UDEP6DATA[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:0]	UDEP6DATA [31:0]	R/W	USB Device Endpoint 6 FIFO Data Port For how many valid reading or writing byte of data is valid, please refer to EP6VB (0xD110018C /0xD2800018C).	

UDEP6VB **0xD110018C/0xD280018C** **Endpoint 6 FIFO Data Port Valid Byte**

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-				UDEP6VB[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:4]	-	-	Reserved	-
[3:0]	UDEP6VB [3:0]	R/W	USB Device Endpoint 6 Data Port Valid Byte UDEP6VB indicates which Byte(s) is valid in UDEP6DATA (0xD1100188/0xD2800188). Data Port [31:0] = {Byte3, Byte2, Byte1, Byte0}.	UDEP6VB = 0: No byte is valid. UDEP6VB[0] = 1: Byte0 is valid. UDEP6VB[1] = 1: Byte1 is valid. UDEP6VB[2] = 1: Byte2 is valid. UDEP6VB[3] = 1: Byte3 is valid.

EP7CTRL 0xD11001C0/0xD28001C0 Endpoint 7 Control

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	INT_CONFIG[3:0]				UDC_INTF_EP7[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	INT_CONFIG_REAL[3:0]				INT_CONFIG_x[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EP7BUF_FLUSH	-			EP7_VLD	-		EP7_EN
Default	0	0	0	0	0	0	0	1

Bit	Function	Type	Description	Condition
[31:24]	-	-	Reserved	-
[23:20]	INT_CONFIG [3:0]	R	EP7 Alter Interface Information The value of INT_CONFIG defines the maximum package size every SOF of Endpoint7 can transfer.	4'h0: The package size is 0. 4'h1: The maximum package size is 16 Byte.

Bit	Function	Type	Description	Condition
			When the host issues a "Set_Interface" command to EP7, the Alter interface of EP7 will be configured. Note: The interface mentioned above is Interface3 of EP5. For details, please refer to Table 1.1.	4'h2: The maximum package size is 32 Byte. 4'h3: The maximum package size is 64 Byte. 4'h4: The maximum package size is 128 Byte. 4'h5: The maximum package size is 256 Byte.
[19:16]	UDC_INTF_EP7 [3:0]	R/W	EP7 Interface Number When the host issues a "Set_Interface" command to EP7, the interface of this EP7 will be configured. Note: The interface mentioned above is Interface7 of EP7. For details, please refer to Table 1.1.	
[15:12]	INT_CONFIG_REAL [3:0]	R	Real Alter Interface Information for USB Controller.	
[11:8]	INT_CONFIG_x[3:0]	R	Not Used	
[7]	EP7BUF_FLUSH	R/W	Endpoint 7 Buffer Flush Set this bit as 1, and it will clear EP7 pointer and buffer. This bit will not be cleared as 0 automatically. User has to write 0 to this bit after flush to stop flush function.	
[6:4]	-	-	Reserved	-
[3]	EP7_VLD	R/W	Endpoint 7 Data Valid When Endpoint 7 IN data is ready (valid), user has to set this bit as 1 to trigger USB Device controller to wait for the IN data of the host. When USB device controller receives the ACK from the host, this bit will be cleared automatically.	0: The data is not valid. 1: The data is valid.
[2:1]	-	-	Reserved	-
[0]	EP7_EN	R/W	Endpoint 7 Enable Bit	0: Disable Endpoint 7. 1: Enable Endpoint 7.

EP7PTRR
0xD11001C4/0xD28001C4
Endpoint 7 FIFO Read Pointer

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							UDEP7PTRR[8]
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	UDEP7PTRR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:9]	-	-	Reserved	-
[8:0]	UDEP7PTRR[8:0]	R/W	USB Device Endpoint 7 FIFO Read Pointer	

EP7PTR 0xD11001C8/0xD28001C8 Endpoint 7 FIFO Write Pointer

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							UDEP7PTR[8]
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	UDEP7PTR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:9]	-	-	Reserved	-
[8:0]	UDEP7PTR[8:0]	R/W	USB Device Endpoint 7 FIFO Write Pointer	

UDEP7DATA	0xD11001CC/0xD28001CC				Endpoint 7 FIFO Data Port			
Bit	31	30	29	28	27	26	25	24
Function	UDEP7DATA[31:24]							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	UDEP7DATA[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	UDEP7DATA[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	UDEP7DATA[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:0]	UDEP7DATA [31:0]	R/W	USB Device Endpoint 7 FIFO Data Port For how many valid reading or writing byte of data is valid, please refer to EP7VB (0xD11001D0/0xD280001D0).	

UDEP7VB	0xD11001D0/0xD28001D0				Endpoint 7 FIFO Data Port Valid Byte			
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-				UDEP7VB[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:4]	-	-	Reserved	-
[3:0]	UDEP7VB [3:0]	R/W	USB Device Endpoint 7 Data Port Valid Byte UDEP7VB indicates which Byte(s) is valid in UDEP7DATA (0xD11001CC/0xD28001CC). Data Port [31:0] = {Byte3, Byte2, Byte1, Byte0}.	UDEP7VB = 0: No byte is valid. UDEP7VB[0] = 1: Byte0 is valid. UDEP7VB[1] = 1: Byte1 is valid. UDEP7VB[2] = 1: Byte2 is valid. UDEP7VB[3] = 1: Byte3 is valid.

EP5STCDIVL 0xD1100200/0xD2800200 EP5 Reference Clock Divisor Low Byte

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	UDSTCDIVL[7:0]							
Default	0	0	1	0	1	0	0	1

Bit	Function	Type	Description	Condition
[31:8]	-	-	Reserved	-
[7:0]	UDSTCDIVL [7:0]	R/W	EP5 Video Class Reference Clock Divisor Low Byte {UDSTCDIVH[7:0], UDSTCDIVL[7:0]} = stclkcnt Period	

EP5STCDIVH		0xD1100204/0xD2800204				EP5 Reference Clock Divisor High Byte		
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit		23	22	21	20	19	18	17	16
Function	-								
Default		0	0	0	0	0	0	0	0

Bit		15	14	13	12	11	10	9	8
Function	-								
Default		0	0	0	0	0	0	0	0

Bit		7	6	5	4	3	2	1	0
Function	UDSTCDIVH[7:0]								
Default		0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:8]	-	-	Reserved	-
[7:0]	UDSTCDIVH [7:0]	R/W	EP5 Reference Clock Divisor High Byte {UDSTCDIVH[7:0], UDSTCDIVL[7:0]} = stclkcnt Period	

EP12C		0xD1100330/0xD2800330				Endpoint 1 and Endpoint 2 Control		
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit		23	22	21	20	19	18	17	16
Function	-								
Default		0	0	0	0	0	0	0	0

Bit		15	14	13	12	11	10	9	8
Function	-					EP2_OUT_ ON	EP1_IN_O N	EP2_OUT_ ENA	EP1_IN_EN A
Default		0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	MSDC_CMD_VLD	EP2_OVLD	EP1_IVLD	SET_EP1_I_VLD	CLR_EP2_O_VLD	RESET_PIP_O_FIFO	-	EP12_ENA_CTRL_CLR
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:12]	-	-	Reserved	-
[11]	EP2_OUT_ON	R/W	EP2_OUT_ON Set EP12C[9] as 1, and this bit will automatically be set as 1. When endpoint 2 Bulk Out DMA transaction is done, this bit will be cleared. In PIO mode, when EP12C[3] = 1, this bit will also be cleared.	0: Disable BULK_OUT_ON. 1: Enable BULK_OUT_ON.
[10]	EP1_IN_ON	R/W	EP1_IN_ON Set EP12C[8] as 1, and this bit will automatically be set as 1. In PIO mode, when all PIO data are piped in FIFO and then be sent out, and the host sends out ACK, this bit will be cleared automatically. In DMA mode, when DMA finishes all the data, this bit will be cleared automatically.	0: Disable BULK_IN_ON. 1: Enable BULK_IN_ON.
[9]	EP2_OUT_ENA	R/W	In PIO/DMA Mode, when USB transaction is done, EP2_OUT_ENA will be cleared automatically. At this time, EP2_OUT_ON will not be cleared. EP2_OUT_ON bit will be cleared when DMA or PIO is done.	0: Disable BULK_OUT. 1: Enable BULK_OUT.
[8]	EP1_IN_ENA	R/W	In PIO/DMA Mode, EP1_IN_ENA will be cleared automatically when all PIO data are piped in FIFO and then be sent out, or when DMA finishes maximum-packet-size data and the host sends out ACK.	0: Disable BULK_IN. 1: Enable BULK_IN.
[7]	MSDC_CMD_VLD	R	Mass Storage Device Class Command Check Valid	
[6]	EP2_OVLD	R	Current Endpoint 2 OUT Buffer Valid Flag Status The function of EP2_OVLD is the same as 0xD1100330/0xD2800330 [5]. It will be cleared when 0xD1100330/0xD2800330[3] are set as 1. In DMA mode, EP2_OVLD will be cleared automatically when DMA is done.	
[5]	EP1_IVLD	R	Current Endpoint 1 IN Buffer Valid Flag Status The function of EP1_IVLD is the same as 0xD1100334/0xD2800334 [4]. It will be set as 1 when 0xD1100330/0xD2800330[4] are set as 1. In DMA mode, EP1_IVLD will be set as 1 automatically.	

Bit	Function	Type	Description	Condition
[4]	SET_EP1_IVLD	W	Set Current endpoint 1 IN Buffer Valid Flag. When Endpoint 1 data is ready, user has to set this bit as 1.	
[3]	CLR_EP2_OVLD	W	Clear Current Endpoint 2 OUT Buffer Valid Flag.	
[2]	RESET_PIPO_FIFO	W	Reset the Current and the Next Endpoint 1 and Endpoint 2 FIFO.	0: No action. 1: Reset the Current and the Next Endpoint 1 and Endpoint 2 FIFO
[1]	-	-	Reserved	-
[0]	EP12_ENA_CTRL_CLR	R/W		0: No function. 1: EP12 control function will be reset, and user has to set it as 0 manually.

EP12PPC
0xD1100334/0xD2800334
Endpoint 1 and Endpoint 2 Ping-Pong FIFO
Control

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	A_EP2_OVLD	A_EP1_IVLD	EP2_OVLD	EP1_IVLD	-	Current_Buffer	Switch_Buffer	Auto_Switch_Enh
Default	0	0	0	0	0	0	0	1

Bit	Function	Type	Description	Condition
[31:8]	-	-	Reserved	-
[7]	A_EP2_OVLD	R	Another Endpoint 2 OUT Buffer Valid Flag Status Which buffer is the another buffer depends on	

Bit	Function	Type	Description	Condition
			0xD1100334/0xD2800334[2]. 0xD1100334/0xD2800334 [2] = 1 → Pong Buffer 0xD1100334/0xD2800334 [2] = 0 → Ping Buffer	
[6]	A_EP1_IVLD	R	Another Endpoint 1 IN Buffer Valid Flag Status Which buffer is the another buffer depends on 0xD1100334/0xD2800334 [2]. 0xD1100334/0xD2800334 [2] = 1 → Pong Buffer 0xD1100334/0xD2800334 [2] = 0 → Ping Buffer	
[5]	EP2_OVLD	R	Current Endpoint 2 OUT Buffer Valid Flag Status Which buffer is the current valid buffer depends on 0xD1100334/0xD2800334 [2]. 0xD1100334/0xD2800334 [2] = 1 → Ping Buffer 0xD1100334/0xD2800334 [2] = 0 → Pong Buffer	
[4]	EP1_IVLD	R	Current Endpoint 1 IN Buffer Valid Flag Status Which buffer is the current valid buffer depends on 0xD1100334/0xD2800334 [2]. 0xD1100334/0xD2800334 [2] = 1 → Ping Buffer 0xD1100334/0xD2800334 [2] = 0 → Pong Buffer	
[3]	-	-	Reserved	-
[2]	Current_Buff	R		0: Buffer1 (PONG) buffer can be controlled by the CPU/DMA. 1: Buffer0 (PING) buffer is controllable for CPU/DMA.
[1]	Switch_Buffer	W	Write 1 to this bit, and user can switch between Buffer0 and Buffer1.	
[0]	Auto_Switch_En	R/W		0: PING PONG buffer will NOT switch automatically. 1: PING PONG buffer will switch automatically.

EP0VB	0xD1100340/0xD2800340	Endpoint 0 Data Port Valid Byte
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Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-				UDEPOVB[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:4]	-	-	Reserved	-
[3:0]	UDEPOVB [3:0]	R/W	USB Device Endpoint 0 Data Port Valid Byte UDEPOVB indicates which Byte(s) is valid in UDEPODATA (0xD1100354/0xD2800354). Data Port [31:0] = {Byte3, Byte2, Byte1, Byte0}.	UDEPOVB = 0: No byte is valid. UDEPOVB[0] = 1: Byte0 is valid. UDEPOVB[1] = 1: Byte1 is valid. UDEPOVB[2] = 1: Byte2 is valid. UDEPOVB[3] = 1: Byte3 is valid.

EPOSETUPCS 0xD1100344/0xD2800344 Endpoint 0 Setup Control Status

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-					EPO_SFIFO _UPDATE	EPO_SFIFO _VALID	-
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:3]	-	-	Reserved	-
[2]	EPO_SFIFO_UPDA TE	RC	EPO_SFIFO_UPDATE will be set as 1 by ULDC when EPO Setup FIFO is updated. It will be cleared when	

Bit	Function	Type	Description	Condition
			read by the CPU. This function is only used in Setup Phase of Setup Transactions. When EPO Data is Up data, this bit will be set as 1. It will be cleared when the CPU read this EPOSETUPCS.	
[1]	EPO_SFIFO_VALID	RC	EPO_SFIFO_VALID will be set as 1 by ULDC when EPO Setup FIFO is valid. It will be cleared when read by the CPU. This function is only used in Setup Phase of Setup Transactions. When the device gets EPO ACK, EPO_SFIFO_VALID will be set as 1. It will be cleared when the CPU read EPOSETUPCS.	
[0]	-	-	Reserved	-

EPOSETUPDP
0xD1100348/0xD2800348
Endpoint 0 Setup Data Port

Bit	31	30	29	28	27	26	25	24
Function	EPO_Setup_Data[31:24]							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	EPO_Setup_Data[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	EPO_Setup_Data[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EPO_Setup_Data[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:0]	EPO_Setup_Data [31:0]	R/W	32Bit/8 Byte Setup Packet Data Port EPO_Setup_Data is only used in Setup Phase of Setup Transactions. The IC reads out Setup Phase Data.	

EPOCS
0xD110034C/0xD280034C
Endpoint 0 Control Status

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							EPO_inout_en
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EPO_OUT_EMPTY	EPO_OVLD	CLR_EPO_OUT_VLD	EPO_IVLD	SET_EPO_IN_VLD	EPO_SFIFO_UPDATE	EPO_SFIFO_VALID	EPO_DIR
Default	0	0	0	0	0	0	0	1

Bit	Function	Type	Description	Condition
[31:9]	-	-	Reserved	-
[8]	EPO_inout_en	R/W	EPO IN/OUT Transfer Enable	0: Disable EPO IN/OUT Transfer. 1: Enable EPO IN/OUT Transfer.
[7]	EPO_OUT_EMPTY	R	EPO_OUT_EMPTY is valid only when EPO_OUT_VLD is 1, so user must check EPO_DOUT_VLD first.	0: CTRL_OUT packet is not a zero length packet. 1: CTRL_OUT packet is a zero length packet.
[6]	EPO_OVLD	R	EPO OUT Buffer Valid Flag Status	0: No data in EPO DATA FIFO. 1: There's valid data in EPO DATA FIFO.
[5]	CLR_EPO_OUT_VLD	R/W	Write 1 to this bit, and EPO_DOUT_VLD (free buffer) will be cleared.	
[4]	EPO_IVLD	R	EPO IN Buffer Valid Flag Status	0: Data in EPO DATA FIFO has been received by the host. 1: Data in EPO DATA FIFO is still Valid and has not been read by the host.
[3]	SET_EPO_IN_VLD	R/W	Write 1 to EPO_DIN_VLD when data in FIFO is valid.	

Bit	Function	Type	Description	Condition
[2]	EPO_SFIFO_UPDATE	R	EPO_SFIFO_UPDATE will be set as 1 by ULDC when EPO Setup FIFO is updated (the difference with UDEPOSCS[2] is that this bit is not Read clear). EPO_SFIFO_UPDATE is only used in Setup Phase of Setup Transactions. The Function is the same as 0xD1100344/0xD2800344 [2]. It will be cleared when user reads 0xD1100344/0xD2800344 register	
[1]	EPO_SFIFO_VALID	R	EPO_SFIFO_VALID will be set as 1 by ULDC when EPO Setup FIFO is valid (the difference with UDEPOSCS[1] is that the bit is not Read clear). EPO_SFIFO_VALID is only used in Setup Phase of Setup Transactions. The function of EPO_SFIFO_VALID is the same as 0xD1100344/0xD2800344[1]. This bit will be cleared when 0xD1100344 and 0xD2800344 register are read.	
[0]	EPO_DIR	R/W	When direction is changed, the EPO_FIFO address pointer will be reset as 0 automatically.	0: CTRL_OUT 1: CTRL_IN

Note:

- For EPO Transactions, it is always enabled. 0xD110034C/0xD280034C [0] decides it's IN or OUT transition.
- 0xD110034C/0xD280034C[1]/[2] are used in Setup Phase of Setup Transactions. 0xD110034C/0xD280034C [7]/[6]/[5]/[4]/[3] are used in Data and Status Phase of Setup Transactions.

EP0DC	0xD1100350/0xD2800350						Endpoint 0 Data Count	
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-	EPO_DATA_CNTR[6:0]						
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:7]	-	-	Reserved	-
[6:0]	EPO_DATA_CNTR [6:0]	R/W	In OUT Phase, read EPO_DATA_CNTR and it's Valid Byte Count. In IN Phase and when IN_VLD = 1, read EPO_DATA_CNTR and it will also be Valid Byte Count. In IN Phase but IN_VLD is not 1, read EPO_DATA_CNTR will be a Write pointer address. The unit of data is Byte.	

EP0DP **0xD1100354/0xD2800354** **Endpoint 0 Data Port**

Bit	31	30	29	28	27	26	25	24
Function	EPO_DATA_PORT[31:24]							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	EPO_DATA_PORT[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	EPO_DATA_PORT[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EPO_DATA_PORT[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:0]	EPO_DATA_PORT [31:0]	R/W	USB Device Endpoint 0 FIFO Data Port For how many valid reading or writing byte of data is valid, please refer to EP0VB (0xD1100340/0xD28000340).	

EP12PICL
0xD1100368/0xD2800368
Endpoint 1 and Endpoint 2 PING FIFO Data
Count Low Byte

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EP12_PING_CNTRL[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:8]	-	-	Reserved	-
[7:0]	EP12_PING_CNTRL[7:0]	R/W	Write: Modify read or write pointer. Read: BULK_OUT = Valid byte Count BULK_IN before SET_IN_VLD = Write pointer. BULK_IN after SET_IN_VLD = Valid Byte Count	

EP12PICH
0xD110036C/0xD280036C
Endpoint 1 and Endpoint 2 PING FIFO Data
Count High Byte

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-					Reset_PIN G_CNTR	EP12_PING_CNTRH [1:0]	
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:3]	-	-	Reserved	-
[2]	Reset_PING_CNT R	R/W	Write 1: BULK_OUT = Move r_ptr to 0. BULK_IN = Move w_ptr to 0.	
[1:0]	EP12_PING_CNT RH[1:0]	R/W	Write: Modify read or write pointer. Read: BULK_OUT = Valid byte Count BULK_IN before SET_IN_VLD = write pointer. BULK_IN after SET_IN_VLD = Valid Byte Count	

EP12FDP 0xD1100370/0xD2800370 Endpoint 1 and Endpoint2 FIFO Data Port

Bit	31	30	29	28	27	26	25	24
Function	EP12_DATA_PORT[31:24]							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	EP12_DATA_PORT[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	EP12_DATA_PORT[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EP12_DATA_PORT[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:0]	EP12_DATA_PORT [31:0]	R/W	USB Device Endpoint 1 and Endpoint 2 FIFO Data Port For how many valid reading or writing byte of data is valid, please refer to EP12VB (0xD1100398/0xD28000398).	

EP3CTRL		0xD1100374/0xD2800374					Endpoint 3 Control	
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit		23	22	21	20	19	18	17	16
Function	-								
Default		0	0	0	0	0	0	0	0

Bit		15	14	13	12	11	10	9	8
Function	-								
Default		0	0	0	0	0	0	0	0

Bit		7	6	5	4	3	2	1	0
Function		-				EP3_VLD	-		EP3_EN
Default		0	0	0	0	0	0	0	1

Bit	Function	Type	Description	Condition
[31:4]	-	-	Reserved	-
[3]	EP3_VLD	R/W	Endpoint 3 Data Valid When Endpoint 3 IN data is ready (valid), user has to set this bit as 1 to trigger USB Device controller to wait for the IN data of the host. When USB device controller receives the ACK from the host, this bit will be cleared automatically.	0: The data is not valid. 1: The data is valid.
[2:1]	-	-	Reserved	-
[0]	EP3_EN	R/W	Endpoint 3 Enable Bit	0: Disable endpoint 3. 1: Enable endpoint 3.

EP3PTR		0xD1100378/0xD2800378					Endpoint 3 FIFO Write Pointer	
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit		23	22	21	20	19	18	17	16
Function	-								
Default		0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	UDEP3PTR[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:8]	-	-	Reserved	-
[7:0]	UDEP3PTR[7:0]	R/W	USB Device Endpoint 3 FIFO Write Pointer	

EP3DATA 0xD110037C/0xD280037C Endpoint 3 FIFO Data Port

Bit	31	30	29	28	27	26	25	24
Function	UDEP3DATA[31:24]							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	UDEP3DATA[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	UDEP3DATA[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	UDEP3DATA[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:0]	UDEP3DATA [31:0]	R/W	USB Device Endpoint 3 FIFO Data Port For how many valid reading or writing byte of data is valid, please refer to EP3VB (0xD1100380/0xD28000380).	

UDEP3VB		0xD1100380/0xD2800380				Endpoint 3 FIFO Data Port Valid Byte		
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit		23	22	21	20	19	18	17	16
Function	-								
Default		0	0	0	0	0	0	0	0

Bit		15	14	13	12	11	10	9	8
Function	-								
Default		0	0	0	0	0	0	0	0

Bit		7	6	5	4	3	2	1	0
Function	-				EP3VB[3:0]				
Default		0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:4]	-	-	Reserved	-
[3:0]	EP3VB [3:0]	R/W	USB Device Endpoint 3 Data Port Valid Byte EP3VB indicates which Byte(s) is valid in UDEP3DATA (0xD110037C/0xD280037C). Data Port [31:0] = {Byte3, Byte2, Byte1, Byte0}.	EP3VB = 0: No byte is valid. EP3VB[0] = 1: Byte0 is valid. EP3VB[1] = 1: Byte1 is valid. EP3VB[2] = 1: Byte2 is valid. EP3VB[3] = 1: Byte3 is valid.

EP0ONAKCN		0xD1100384/0xD2800384					Endpoint 0 OUT NAK Count	
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit		23	22	21	20	19	18	17	16
Function	-								
Default		0	0	0	0	0	0	0	0

Bit		15	14	13	12	11	10	9	8
Function	-			EPO_OUT_NAK_CNT[13:8]					
Default		0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EP0_OUT_NAK_CNT[7:0]/EP0_ONAK_THRESHOLD[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:14]	-	-	Reserved	-
[13:8]	EP0_OUT_NAK_CNT[13:8]	R	Read EP0_OUT_NAK_CNT and user can obtain the value of EP0_OUT_NAK_CNT[13:0]. EP0_OUT_NAK_CNT is 14bits. When EP0_OUT_NAK_CNT[13:6] == EP0_ONAK_THRESHOLD[7:0], UDC_IRQ[9] (0xD1100400/0xD2800400) will be triggered.	
[7:0]	EP0_OUT_NAK_CNT[7:0]/EP0_ONAK_THRESHOLD[7:0]	R/W	Write a suitable value to EP0_ONAK_THRESHOLD to determine the interrupt will happen after how many NAK counts. For reading EP0_OUT_NAK_CNT[7:0], please refer to the description in EP0_OUT_NAK_CNT[13:8].	

EP0INAKCN 0xD1100388/0xD2800388 Endpoint 0 IN NAK Count

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-		EP0_IN_NAK_CNT[13:8]					
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EP0_IN_NAK_CNT[7:0]/EP0_INAK_THRESHOLD[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:14]	-	-	Reserved	-
[13:8]	EP0_IN_NAK_CNT[13:8]	R	Read EP0_IN_NAK_CNT and user can obtain the value of EP0_IN_NAK_CNT[13:0]. EP0_IN_NAK_CNT is 14bits. When EP0_IN_NAK_CNT[13:6] == EP0_INAK_THRESHOLD[7:0], UDC_IRQ[9] (0xD1100400/0xD2800400) will be triggered.	

Bit	Function	Type	Description	Condition
[7:0]	EP0_IN_NAK_CNT[7:0]/EP0_IN_NAK_THRESHOLD[7:0]	R/W	Write a suitable value to EP0_INAK_THRESHOLD to determine the interrupt will happen after how many NAK counts. For reading EP0_INAK_THRESHOLD[7:0], please refer to the description in EP0_IN_NAK_CNT[13:8].	

EP1INAKCN 0xD110038C/0xD280038C Endpoint 1 IN NAK Count

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-				EP1_INAK_CNT[13:8]			
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EP1_INAK_CNT[7:0]/EP1_INAK_THRESHOLD[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:14]	-	-	Reserved	-
[13:8]	EP1_INAK_CNT[13:8]	R	Read EP1_INAK_CNT and user can obtain the value of EP1_INAK_CNT[13:0]. EP1_INAK_CNT is 14bits. When EP1_INAK_CNT[13:6] == EP1_INAK_THRESHOLD[7:0], UDC_IRQ[10] (0xD1100400/0xD2800400) will be triggered.	
[7:0]	EP1_INAK_CNT[7:0]/EP1_INAK_THRESHOLD[7:0]	R/W	Write a suitable value to EP1_INAK_THRESHOLD to determine the interrupt will happen after how many NAK counts. For reading EP1_INAK_THRESHOLD[7:0], please refer to the description in EP1_INAK_CNT[13:8].	

EP2ONAKCN 0xD1100390/0xD2800390 Endpoint 2 OUT NAK Count

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-		EP2_ONAK_CNT[13:8]					
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EP2_ONAK_CNT[7:0]/EP2_ONAK_THRESHOLD[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:14]	-	-	Reserved	-
[13:8]	EP2_ONAK_CNT T[13:8]	R	Read EP2_ONAK_CNT and user can obtain the value of EP2_ONAK_CNT[13:0]. EP2_ONAK_CNT is 14bits. When EP2_ONAK_CNT[13:6] == EP2_ONAK_THRESHOLD[7:0], UDC_IRQ[11] (0xD1100400/0xD2800400) will be triggered.	
[7:0]	EP2_ONAK_CNT T[7:0]/EP2_ONAK_THRESHOLD [7:0]	R/W	Write a suitable value to EP2_ONAK_THRESHOLD to determine the interrupt will happen after how many NAK counts. For reading EP2_ONAK_CNT[7:0], please refer to the description in EP2_ONAK_CNT[13:8].	

UDEP12VB 0xD1100398/0xD2800398 Endpoint 1 and Endpoint 2 FIFO Data Port Valid
Byte

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-				UDEP12VB[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:4]	-	-	Reserved	-
[3:0]	EP12VB [3:0]	R/W	USB Device Endpoint 12 Data Port Valid Byte EP12VB indicates which Byte(s) is valid in EP12DATA (0xD1100370/0xD2800370). Data Port [31:0] = {Byte3, Byte2, Byte1, Byte0}.	EP12VB = 0: No byte is valid. EP12VB[0] = 1: Byte0 is valid. EP12VB[1] = 1: Byte1 is valid. EP12VB[2] = 1: Byte2 is valid. EP12VB[3] = 1: Byte3 is valid.

EP12POCL **0xD110039C/0xD280039C** **Endpoint 1 and Endpoint 2 PONG FIFO Data**
Count Low Byte

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EP12_PONG_CNTRL[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:8]	-	-	Reserved	-
[7:0]	EP12_PONG_CNTRL[7:0]	R/W	Write: Modify read or write pointer. Read: BULK_OUT = Valid Byte Count BULK_IN before SET_IN_VLD = Write pointer. BULK_IN after SET_IN_VLD = Valid Byte Count	

EP12POCH
0xD11003A0/0xD28003A0
Endpoint 1 and Endpoint 2 PONG FIFO Data
Count High Byte

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-					Reset_PONG_CNTR	EP12_PONG_CNTRH[1:0]	
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:3]	-	-	Reserved	-
[2]	Reset_PONG_CNTR	R/W	When this bit is set as 1, EP12_PONG_CNTRH will be reset. But this bit can only be cleared by writing 0 manually.	
[1:0]	EP12_PONG_CNTRH[1:0]	R/W	Write: Modify read or write pointer. Read: BULK_OUT = Valid byte Count BULK_IN before SET_IN_VLD = Write pointer. BULK_IN after SET_IN_VLD = Valid Byte Count	

UDLCSET0
0xD11003B0/0xD28003B0
USB Device Linker Layer Controller Setting0

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	Clear_Suspend_Counter	SIM_Mode	Disconnect_Suspend_En	CPU_WakeUp_En	PWR_Partial_n	PWR_Suspend_n	Issue_Resume	dev_Disconnect
Default	0	0	0	0	1	1	0	1

Bit	Function	Type	Description	Condition
[31:8]	-	-	Reserved	-
[7]	Clear_Suspend_Counter	R/W		0: No action. 1: Force UDC to suspend counter reset.
[6]	SIM_Mode	R/W		0: Normal. 1: Increase simulation speed by reducing counter threshold.
[5]	Disconnect_Suspend_En	R/W		0: UDC linker will not request USB-PHY to suspend when the device is disconnected to the host. 1: When the device is disconnected to the host, UDC linker will request USB-PHY to suspend.
[4]	CPU_WakeUp_En	R/W		
[3]	PWR_Partial_n	R/W		0: No USB-PHY clock when the device suspends. 1: Keep USB-PHY clock when the device suspends.
[2]	PWR_Suspend_n	R/W		0: Allow LNK (udc linker) to suspend PHY. 1: Prevent LNK to suspend PHY.
[1]	Issue_Resume	R/W		0: Normal 1: The device will resume, and this bit will be cleared as 0 automatically.
[0]	dev_Disconnect	R/W		0: DP is pulled high, and the device is in the state that's ready for connecting. 1: DP and DM are pulled low, and the device is in the state of disconnecting.

UDLCSET1 0xD11003B4/0xD28003B4 USB Device Linker Layer Controller Setting1

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-	NO_STOP CHIRP	Inter_Pack et_Delay	FORCE_FU LLSP	VBUS_LOW _AUTO_DIS C	DISC_AUT O_DPDMP D	SUPP_RWA KE	SELF_POW ER
Default	0	0	0	0	0	0	0	1

Bit	Function	Type	Description	Condition
[31:7]	-	-	Reserved	-
[6]	NO_STOP CHIRP	R/W		0: Stop chirp automatically when Full speed J is high. 1: Disable this function.
[5]	Inter_Packet_Delay	R/W		0: Normal delay. 1: One additional delay cycle.
[4]	FORCE_FULLSP	R/W		0: Normal. 1: Force the device to be in full-speed state.
[3]	VBUS_LOW_AUTO _DISC	R/W		0: Normal. 1: The device will be disconnected to the host automatically when VBUS is absent.
[2]	DISC_AUTO_DPDMP PD	R/W	Pin Serial Mode DP/DM Pull Down Selection When the device is disconnected to the host, set this bit as 1, and Pin Serial mode DP/DM will be pulled down.	0: Normal. DP/DM will not be pulled down when CONNECT_STATUS is 0. 1: Pull down DP/DM automatically when CONNECT_STATUS is 0.
[1]	Remote_RWAKE	R/W		0: Set the IC as NOT supporting the "remote wakeup" function. 1: Set the IC as supporting the "remote wake up" function.
[0]	SELF_POWER	R/W		0: Set the IC as a bus power device. 1: Set the IC as a self-power device.

UDLCS		0xD11003B8/0xD28003B8				USB Device Linker Layer Controller Status		
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	DISC_CONNE CT_STATUS	DEV_CO NFIG	LNK_SUSPE NDM	H_ALLOW_ RWAKE	CURR_LINSTATE[1:0]		CURR_SPE ED	VBUS_HIG H
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:8]	-	-	Reserved	-
[7]	DISC_CONNECT_STA TUS	R	Device Disconnect	0: UDC DEVICE is connected. 1: UDC DEVICE is disconnected.
[6]	DEV_CONFIG	R	UDC Configuration Value	
[5]	LNK_SUSPENDM	R	Device Go suspend	0: LNK tries to suspend PHY. 1: Normal.
[4]	ALLOW_RWAKE	R	Device Support Remote Wake-Up	0: The host does not allow remote wake-up. 1: The host allows remote wake-up.
[3:2]	CURR_LINSTATE [1:0]	R	Device Line State.	Full Speed: 00 = SE0 01 = J 10 = K High Speed: 00 = SE0 01 = Squelch Reset Data is invalid.
[1]	CURR_SPEED	R	Device Speed.	0: High speed. 1: Full speed.
[0]	VBUS_HIGH	R	Device VBUS Value	0: VBUS is detected as low. 1: VBUS is detected as high.

UDLCSTL		0xD11003BC/0xD28003BC				USB Device Linker Layer Controller Stall Control		
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	CLREPBSTL	CLREPASTL	CLREP9STL	CLREP8STL	CLREP3STL	CLREP2STL	CLREP1STL	CLREPOSTL
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	SETEPBSTL	SETEPASTL	SETEP9STL	SETEP8STL	SETEP3STL	SETEP2STL	SETEP1STL	SETEPOSTL
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:16]	-	-	Reserved	-
[15]	CLREPBSTL	R/W	Clear endpoint 11 stall.	
[14]	CLREPASTL	R/W	Clear endpoint 10 stall.	
[13]	CLREP9STL	R/W	Clear endpoint 9 stall.	
[12]	CLREP8STL	R/W	Clear endpoint 8 stall.	
[11]	CLREP3STL	R/W	Clear endpoint 3 stall.	
[10]	CLREP2STL	R/W	Clear endpoint 2 stall.	
[9]	CLREP1STL	R/W	Clear endpoint 1 stall.	
[8]	CLREPOSTL	R/W	Clear endpoint 0 stall.	
[7]	SETEPBSTL	R/W	Set endpoint 11 stall.	
[6]	SETEPASTL	R/W	Set endpoint 10 stall.	
[5]	SETEP9STL	R/W	Set endpoint 9 stall.	
[4]	SETEP8STL	R/W	Set endpoint 8 stall.	
[3]	SETEP3STL	R/W	Set endpoint 3 stall.	
[2]	SETEP2STL	R/W	Set endpoint 2 stall.	
[1]	SETEP1STL	R/W	Set endpoint 1 stall.	
[0]	SETEPOSTL	R/W	Set endpoint 0 stall.	

UDLCSET2 0xD11003C0/0xD28003C0 USB Device Linker Layer Controller Setting2

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-							SUP_EP3
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:1]	-	-	Reserved	-
[0]	SUP_EP3	R/W	Support EP3 Function	0: Not support endpoint 3. 1: Support endpoint 3.

UDLCS2

0xD11003C4/0xD28003C4

USB Device Linker Layer Controller Status2

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-				Full_LS[1:0]		Sync_LS[1:0]	
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:4]	-	-	Reserved	-
[3:2]	Full_LS[1:0]	R	UTMI Serial Mode(Full Speed) Line State(dm/dp)	
[1:0]	Sync_LS[1:0]	R	UTMI Interface Line State(linestate[1:0])	

UDLC_INTERFACE	0xD11003C8/0xD28003C8				USB Device Linker Layer Controller Interface			
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-			Interface_set[4:0]				
Default	0	0	0	1	1	1	1	1

Bit	Function	Type	Description	Condition
[31:5]	-	-	Reserved	-
[4:0]	Interface_set [4:0]	R/W	Pile Enable Bit For setting endpoint interface number, please refer to Table 1.1. Interface1: Endpoint 1 (Bulk in), Endpoint 2 (Bulk Out), Endpoint 3 (INT IN) Interface Number Interface2: Endpoint 4 (INT IN) Interface Number Interface3: Endpoint 5 (Video IOS IN) Interface Number Interface4: Endpoint 6 (INT IN) Interface Number Interface5: Endpoint 7 (isochronous in for audio) Interface Number	Bit 0: MSC Enable Bit 1: Video Class Interrupt Pipe Enable Bit 2: Video Class Data Pipe Enable Bit 3: Audio Class Interrupt Pipe Enable Bit 4: Audio Class Data Pipe Enable

Interface 1~5 Map Table

Interface_set[4:0]	Interface1[3:0]	Interface2[3:0]	Interface3[3:0]	Interface4[3:0]	Interface5[3:0]
5'h00000	4	4	4	4	4
5'h00001	0	4	4	4	4
5'h00010	4	0	4	4	4
5'h00011	0	1	4	4	4
5'h00100	4	4	0	4	4
5'h00101	0	4	1	4	4
5'h00110	4	0	1	4	4

Interface_set[4:0]	Interface1[3:0]	Interface2[3:0]	Interface3[3:0]	Interface4[3:0]	Interface5[3:0]
5'h00111	0	1	2	4	4
5'h01000	4	4	4	0	4
5'h01001	0	4	4	1	4
5'h01010	4	0	4	1	4
5'h01011	0	1	4	2	4
5'h01100	4	4	0	1	4
5'h01101	0	4	1	2	4
5'h01110	4	0	1	2	4
5'h01111	0	1	2	3	4
5'h10000	4	4	4	4	0
5'h10001	0	4	4	4	1
5'h10010	4	0	4	4	1
5'h10011	0	1	4	4	2
5'h10100	4	4	0	4	1
5'h10101	0	4	1	4	2
5'h10110	4	0	1	4	2
5'h10111	0	1	2	4	3
5'h11000	4	4	4	0	1
5'h11001	0	4	4	1	2
5'h11010	4	0	4	1	2
5'h11011	0	1	4	2	3
5'h11100	4	4	0	1	2
5'h11101	0	4	1	2	3
5'h11110	4	0	1	2	3
5'h11111(Default)	0	1	2	3	4

UDLCADDR	0xD11003F4/0xD28003F4						Device Address Register	
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	ADDR_VLD	DEV_ADDR[6:0]						
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:8]	-	-	Reserved	-
[7]	ADDR_VLD	R	Address Valid Flag	
[6:0]	DEV_ADDR[6:0]	R	Device Address	

UDC_IF	0xD1100400/0xD2800400						USB Device Linker Layer Interrupt Flag	
Bit	31	30	29	28	27	26	25	24
Function	-						EP7_DMA_IF	EP5_DMA_IF
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	EP7I_IF	EP6I_IF	EP5I_IF	EP4I_IF	URES_IF	HSC_IF	RESU_IF	SUS_IF
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	EP1_DMA_IF	EP3I_IF	PIPO_IF	HCS_IF	EP2N_IF	EP1N_IF	EP0N_IF	HSS_IF
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EP2O_IF	EP1I_IF	EP1SI_IF	EP0I_IF	EP0O_IF	EP0S_IF	SUSP_IF	RESET_IF
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:26]	-	-	Reserved	-
[25]	EP7_DMA_IF	R/W	Endpoint 7 DMA IN Data Finish Interrupt Flag EP7_DMA_IF will be set as 1 when EP7 DMA function finishes and the device gets EP7 IN ack. Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[24]	EP5_DMA_IF	R/W	Endpoint 5 DMA IN Data Finish Interrupt Flag EP5_DMA_IF will be set as 1 when EP5 DMA function finishes and the device gets IN EP5 ack. Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.

Bit	Function	Type	Description	Condition
[23]	EP7I_IF	R/W	Endpoint 7 IN Transaction Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[22]	EP6I_IF	R/W	Endpoint 6 IN Transaction Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[21]	EP5I_IF	R/W	Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[20]	EP4I_IF	R/W	Endpoint 4 IN Transaction Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[19]	URES_IF	R/W	USB_RESET Release Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[18]	HSC_IF	R/W	Host Set Configuration Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[17]	RESU_IF	R/W	Resume Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[16]	SUS_IF	R/W	Suspend Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[15]	EP1_DMA_IF	R/W	Endpoint 1 DMA IN Finish Interrupt Flag It will set 1 when DMA function finishes and device gets IN ack. Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[14]	EP3I_IF	R/W	Endpoint 3 IN Transaction Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[13]	PIPO_IF	R/W	Endpoint 1 and Endpoint 2 PING-PONG_FIFO_SWAP Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[12]	HCS_IF	R/W	Host CLEAR_STALL Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[11]	EP2N_IF	R/W	Endpoint 2 NAK Transaction Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[10]	EP1N_IF	R/W	Endpoint 1 NAK Transaction Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[9]	EPON_IF	R/W	Endpoint 0 IN/OUT NAK Transaction Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[8]	HSS_IF	R/W	Host SET_STALL Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[7]	EP2O_IF	R/W	Endpoint 2 OUT Transaction Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[6]	EP1I_IF	R/W	Endpoint 1 IN Transaction Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[5]	EP1SI_IF	R/W	Endpoint 1S IN Transaction Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[4]	EP0I_IF	R/W	Endpoint 0 IN Transaction Interrupt Flag	0: Interrupt is not asserted.

Bit	Function	Type	Description	Condition
			Write 1 to clear this bit.	1: Interrupt is asserted.
[3]	EPOO_IF	R/W	Endpoint 0 OUT Transaction Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[2]	EPOS_IF	R/W	Endpoint 0 SETUP Transaction Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[1]	SUSP_IF	R/W	Suspend Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.
[0]	RESET_IF	R/W	RESET Interrupt Flag Write 1 to clear this bit.	0: Interrupt is not asserted. 1: Interrupt is asserted.

UDC_IE 0xD1100404/0xD2800404 USB Device Linker Layer Interrupt Enable

Bit	31	30	29	28	27	26	25	24
Function	-						EP7_DMA_ IE	EP5_DMA_ IE
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	EP7I_IE	EP6I_IE	EP5I_IE	EP4I_IE	URES_IE	HSC_IE	RESU_IE	SUS_IE
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	EP12_DMA_ _IE	EP3I_IE	PIPO_IE	HCS_IE	EP2N_IE	EP1N_IE	EPON_IE	HSS_IE
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EP2O_IE	EP1I_IE	EP1SI_IE	EPOI_IE	EPOO_IE	EPOS_IE	SUSP_IE	RESET_IE
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:26]	-	-	Reserved	-
[25]	EP7_DMA_IE	R/W	Endpoint 7 DMA IN Data Finish Interrupt Enable	0: Disable 1: Enable
[24]	EP5_DMA_IE	R/W	Endpoint 5 DMA IN Data Finish Interrupt Enable	0: Disable 1: Enable
[23]	EP7I_IE	R/W	EP7_IN Transaction Interrupt Enable	0: Disable 1: Enable
[22]	EP6I_IE	R/W	EP6_IN Transaction Interrupt Enable	0: Disable

Bit	Function	Type	Description	Condition
				1: Enable
[21]	EP5I_IE	R/W	EP5_IN Transaction Interrupt Enable	0: Disable 1: Enable
[20]	EP4I_IE	R/W	EP4_IN Transaction Interrupt Enable	0: Disable 1: Enable
[19]	URES_IE	R/W	USB_RESET Release Interrupt Enable	0: Disable 1: Enable
[18]	HSC_IE	R/W	Host Set Configuration Interrupt Enable	0: Disable 1: Enable
[17]	RESU_IE	R/W	Resume Interrupt Enable	0: Disable 1: Enable
[16]	SUS_IE	R/W	Suspend Interrupt Enable	0: Disable 1: Enable
[15]	EP12_DMA_IE	R/W	Endpoint 1 and Endpoint 2 DMA IN Data Finish Interrupt Enable	0: Disable 1: Enable
[14]	EP3I_IE	R/W	EP3_IN Transaction Interrupt Enable	0: Disable 1: Enable
[13]	PIPO_IE	R/W	Endpoint 1 and Endpoint 2 PING-PONG_FIFO_SWAP Interrupt Enable	0: Disable 1: Enable
[12]	HCS_IE	R/W	HOST_CLEAR_STALL Interrupt Enable	0: Disable 1: Enable
[11]	EP2N_IE	R/W	EP2_NAK Interrupt Enable	0: Disable 1: Enable
[10]	EP1N_IE	R/W	EP1_NAK Interrupt Enable	0: Disable 1: Enable
[9]	EP0N_IE	R/W	EP0_NAK Interrupt Enable	0: Disable 1: Enable
[8]	HSS_IE	R/W	HOST_SET_STALL Interrupt Enable	0: Disable 1: Enable
[7]	EP2O_IE	R/W	EP2_OUT Transaction Interrupt Enable	0: Disable 1: Enable
[6]	EP1I_IE	R/W	UDLC EP1_IN Transaction Interrupt Enable	0: Disable 1: Enable
[5]	EP1SI_IE	R/W	EP1S_IN Transaction Interrupt Enable	0: Disable 1: Enable
[4]	EP0I_IE	R/W	EP0_IN Transaction Interrupt Enable	0: Disable 1: Enable
[3]	EP0O_IE	R/W	EP0_OUT Transaction Interrupt Enable	0: Disable 1: Enable
[2]	EP0S_IE	R/W	EP0_SETUP Transaction Interrupt Enable	0: Disable 1: Enable
[1]	SUSP_IE	R/W	USB_SUSPEND_DIFF Interrupt Enable	0: Disable 1: Enable
[0]	RESET_IE	R/W	USB_RESET Interrupt Enable	0: Disable 1: Enable

USB_STD_IF	0xD1100410/0xD2800410				USB Standard Request Interrupt Flag			
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-	SET_DESC_IF	GET_DESC_IF	GET_CONF_IF	GET_STS_IF	GET_INTF_IF	SET_INTF_IF	SET_ADDR_IF
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	SET_CONF_IF	SET_TEST_IF	SET_REMOTE_IF	SET_EPx_STALL_IF	SET_EP0_STALL_IF	CLR_REMOTE_IF	CLR_EPx_STALL_IF	CLR_EP0_STALL_IF
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:15]	-	-	Reserved	-
[14]	SET_DESC_IF	R/W	Set-Descriptor Command Interrupt Flag	
[13]	GET_DESC_IF	R/W	Get-Descriptor Interrupt Flag	
[12]	GET_CONF_IF	R/W	Get-Configuration Interrupt Flag	
[11]	GET_STS_IF	R/W	Get-Status Interrupt Flag	
[10]	GET_INTF_IF	R/W	Get-Interface Interrupt Flag	
[9]	SET_INTF_IF	R/W	Set-Interface Interrupt Flag	
[8]	SET_ADDR_IF	R/W	Set-Address Interrupt Flag	
[7]	SET_CONF_IF	R/W	Set-Configuration Interrupt Flag	
[6]	SET_TEST_IF	R/W	Set-Feature (test_mode) Interrupt Flag	
[5]	SET_REMOTE_IF	R/W	Set-Feature Interrupt Flag (device_remote_wakeup)	
[4]	SET_EPx_STALL_IF	R/W	Set-Feature Interrupt Flag for Other EP Stall	
[3]	SET_EP0_STALL_IF	R/W	Set-Feature Command Interrupt Flag for EP0 Stall	
[2]	CLR_REMOTE_IF	R/W	Clear-Feature Interrupt Flag (device_remote_wakeup)	
[1]	CLR_EPx_STALL_IF	R/W	Clear-Feature Interrupt Flag for Other EP Stall	
[0]	CLR_EP0_STALL_IF	R/W	Clear-Feature Interrupt Flag for EP0 Stall	

USB_STD_IE	0xD1100414/0xD2800414				USB Standard Request Interrupt Enable			
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-	SET_DESC_IE	GET_DESC_IE	GET_CONF_IE	GET_STS_IE	GET_INTF_IE	SET_INTF_IE	SET_ADDR_IE
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	SET_CONF_IE	SET_TEST_IE	SET_REMOTE_IE	SET_EPx_STALL_IE	SET_EP0_STALL_IE	CLR_REMOTE_IE	CLR_EPx_STALL_IE	CLR_EP0_STALL_IE
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:15]	-	-	Reserved	-
[14]	SET_DESC_IE	R/W	Set-Descriptor Interrupt Enable	
[13]	GET_DESC_IE	R/W	Get-Descriptor Interrupt Enable	
[12]	GET_CONF_IE	R/W	Get-Configuration Interrupt Enable	
[11]	GET_STS_IE	R/W	Get-Status Interrupt Enable	
[10]	GET_INTF_IE	R/W	Get-Interface Interrupt Enable	
[9]	SET_INTF_IE	R/W	Set-Interface Interrupt Enable	
[8]	SET_ADDR_IE	R/W	Set-Address Interrupt Enable	
[7]	SET_CONF_IE	R/W	Set-Configuration Interrupt Enable	
[6]	SET_TEST_IE	R/W	Set-Feature (test_mode) Interrupt Enable	
[5]	SET_REMOTE_IE	R/W	Set-Feature Interrupt Enable (device_remote_wakeup)	
[4]	SET_EPx_STALL_IE	R/W	Set-Feature Interrupt Enable for Other EP Stall	
[3]	SET_EP0_STALL_IE	R/W	Set-Feature Interrupt Enable for EP0 Stall	
[2]	CLR_REMOTE_IE	R/W	Clear-Feature Interrupt Enable (device_remote_wakeup)	
[1]	CLR_EPx_STALL_IE	R/W	Clear-Feature Interrupt Enable for Other EP Stall	
[0]	CLR_EP0_STALL_IE	R/W	Clear-Feature Interrupt Enable for EP0 Stall	

UDC_NEW_IF	0xD1100420/0xD2800420				USB Device New Endpoint Interrupt Flag			
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-	EP12_NAK_OVER	EP12_USB_Fail	EP12_USB_SUCC	EP11_USB_SUCC_IF	EP10_USB_SUCC_IF	EP11_ONAK_OVER_IF	EP10_INAK_OVER_IF
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EP1011_PIPPO_IF	EP1011DMA_A_IF	EP9_USB_SUCC_IF	EP8_USB_SUCC_IF	EP9_ONAK_OVER_IF	EP8_I_NAK_OVER_IF	EP89_PIPPO_IF	EP89DMA_IF
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:15]	-	-	Reserved	-
[14]	EP12_NAK_OVER_IF	R/W	Endpoint 12 Isochronous OUT NAK	
[13]	EP12_USB_Fail_IF	R/W	Endpoint 12 Isochronous OUT Transaction Fail	
[12]	EP12_USB_SUCC_IF	R/W	Endpoint 12 Isochronous OUT Transaction Finish	
[11]	EP11_USB_SUCC_IF	R/W	EP11_OUT Transaction Interrupt Flag	
[10]	EP10_USB_SUCC_IF	R/W	EP10_IN Transaction Interrupt Flag	
[9]	EP11_ONAK_OVER_IF	R/W	EP11_NAK Interrupt Flag	
[8]	EP10_INAK_OVER_IF	R/W	EP10_NAK Interrupt Flag	
[7]	EP1011_PIPPO_IF	R/W	Endpoint 10 and Endpoint 11 PING-PONG_FIFO_SWAP Interrupt Flag	
[6]	EP1011DMA_IF	R/W	Endpoint 10 and Endpoint 11 DMA Data Finish Interrupt Flag	
[5]	EP9_USB_SUCC_IF	R/W	EP9_OUT success Transaction Interrupt Flag	
[4]	EP8_USB_SUCC_IF	R/W	EP8_IN success Transaction Interrupt Flag	
[3]	EP9_ONAK_OVER_IF	R/W	EP9 OUT NAK over Interrupt Flag	
[2]	EP8_I_NAK_OVER_IF	R/W	EP8_IN NAK over Interrupt Flag	
[1]	EP89_PIPPO_IF	R/W	Endpoint 8 and Endpoint 9 PING-PONG_FIFO_SWAP	

Bit	Function	Type	Description	Condition
			Interrupt Flag	
[0]	EP89DMA_IF	R/W	Endpoint 8 and Endpoint 9DMA Data Finish Interrupt Flag	

UDC_NEW_IE 0xD1100424/0xD2800424 USB Device New Endpoint Interrupt Enable

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-	EPC_USB_NAK_OVER_IE	EPC_USB_Fail_IE	EPC_USB_SUCC_IE	EP11O_IE	EP10I_IE	EP11N_IE	EP10N_IE
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	PIPOEPAB_IE	EPADMA_IE	EP9O_IE	EP8I_IE	EP9N_IE	EP8N_IE	EP89PIPO_IE	EP8DMA_IE
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:15]	-	-	Reserved	-
[14]	EP12_NAK_OVER_IE	R/W	Endpoint 12 Isochronous OUT NAK Interrupt Enable	
[13]	EP12_USB_Fail_IE	R/W	Endpoint 12 Isochronous OUT Transaction Fail Interrupt Enable	
[12]	EP12_USB_SUCC_IE	R/W	Endpoint 12 Isochronous OUT Success Transaction Finish Interrupt Enable	
[11]	EP11_USB_SUCC_IE	R/W	EP11_OUT Success Transaction Interrupt Enable	
[10]	EP10_USB_SUCC_IE	R/W	EP10_IN Transaction Interrupt Enable	
[9]	EP11_ONAK_OVER_IE	R/W	EP11 Out NAK Over Interrupt Enable	
[8]	EP10_INAK_OVER_IE	R/W	EP10 IN NAK Over Interrupt Enable	
[7]	EP1011_PIPO_IE	R/W	Endpoint 10 and Endpoint 11 PING-PONG_FIFO_SWAP Interrupt Enable	
[6]	EP1011DMA_IE	R/W	UDLC Endpoint 10 and Endpoint 11 DMA Data Finish Interrupt Enable	

Bit	Function	Type	Description	Condition
[5]	EP9_USB_SUCC_IE	R/W	EP9_OUT Success Transaction Interrupt Enable	
[4]	EP8_USB_SUCC_IE	R/W	EP8_IN Success Transaction Interrupt Enable	
[3]	EP9_ONAK_OVER_IE	R/W	EP9_NAK Interrupt Enable	
[2]	EP8_I_NAK_OVER_IE	R/W	EP8_NAK Interrupt Enable	
[1]	EP89_PIPO_IE	R/W	Endpoint 8 and Endpoint 9 PING-PONG_FIFO_SWAP Interrupt Enable	
[0]	EP89DMA_IE	R/W	Endpoint 8 and Endpoint 9 DMA Data Finish Interrupt Enable	

EP89C
0xD1100500/0xD2800500
Endpoint 8 and Endpoint 9 Control

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-				EP9_OUT_ON	EP8_IN_ON	EP9_OUT_ENA	EP8_IN_ENA
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-	EP9_OVLD	EP8_IVLD	SET_EP8_IVLD	CLR_EP9_OVLD	RESET_PIP_O_FIFO	-	EP89_ENA_CTRL_CLR
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:12]	-	-	Reserved	-
[11]	EP9_OUT_ON	R/W	EP9_OUT_ON Set EP89C[9] as 1, and this bit will automatically be set as 1. When endpoint 9 Bulk Out DMA transaction is done, this bit will be cleared. In PIO Mode, when EP89C[3] = 1, this bit will also be cleared.	0: Disable BULK_OUT_ON. 1: Enable BULK_OUT_ON.
[10]	EP8_IN_ON	R/W	EP8_IN_ON Set EP89C[8] as 1, and this bit will automatically be set as 1. When DMA done or PIO is done, this bit will	0: Disable BULK_IN_ON. 1: Enable BULK_IN_ON.

Bit	Function	Type	Description	Condition
			be cleared automatically.	
[9]	EP9_OUT_ENA	R/W	EP9_OUT Enable In PIO/DMA Mode, when USB transaction is done, EP9_OUT_ENA will be cleared automatically. At this time, EP9_OUT_ON will not be cleared. EP9_OUT_ON Bit will be cleared when DMA or PIO is done.	0: Disable BULK_OUT. 1: Enable BULK_OUT.
[8]	EP8_IN_ENA	R/W	EP8_IN Enable In PIO/DMA Mode, EP8_IN_ENA will be cleared automatically when DMA function finishes and the host sends out ACK.	0: Disable BULK_IN. 1: Enable BULK_IN.
[7]	-	-	Reserved	-
[6]	EP9_OVLD	R	Current Endpoint 9 OUT Buffer Valid Flag Status This bit is the same as EP89C [5], and it will be cleared when EP89C[3] is set as 1. In DMA mode, EP9_OVLD will be cleared automatically when DMA is done.	
[5]	EP8_IVLD	R	Current Endpoint 8 IN Buffer Valid Flag Status This bit is the same as EP89PPC (0xD1100334/0xD2800334) [4]. EP8_IVLD will be set as 1 when EP89C [4] is set as 1. In DMA mode, it will be set as 1 automatically.	
[4]	SET_EP8_IVLD	W	Set Current endpoint 8 IN Buffer Valid Flag. Set this bit as 1 when endpoint 8 Data is ready.	
[3]	CLR_EP9_OVLD	W	Write 1 to CLR_EP9_OVLD, and EP9_OVLD will be cleared.	
[2]	RESET_PIPO_FIFO	W	Set RESET_PIPO_FIFO as 1, the current and the next endpoint 8 and endpoint 9 FIFO will be reset.	0: No action. 1: Reset the current and the next endpoint 8 and endpoint 9 FIFO.
[1]	-	-	Reserved	-
[0]	EP89_ENA_CTRL_CLR	R/W	Clear EP89 Control Register	0: No action. 1: Clear EP89 control function.

EP89PPC		0xD1100504/0xD2800504				Endpoint 8 and Endpoint 9 Ping-Pong Control			
Bit		31	30	29	28	27	26	25	24
Function	-								
Default		0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	A_EP9_OVLD	A_EP8_IVLD	EP9_OVLD	EP8_IVLD	-	Current_Buffer	Switch_Buffer	Auto_Switch_En
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:8]	-	-	Reserved	-
[7]	A_EP9_OVLD	R	Another Endpoint 9 OUT Buffer Valid Flag Status Which buffer is the another buffer depends on 0xD1100504/0xD2800504 [2]. 0xD1100504/0xD2800504 [2] = 1 → Pong Buffer 0xD1100504/0xD2800504 [2] = 0 → Ping Buffer	
[6]	A_EP8_IVLD	R	Another Endpoint 8 IN Buffer Valid Flag Status Which buffer is the another_buffer depends on 0xD1100504/0xD2800504 [2]. 0xD1100504/0xD2800504 [2] = 1 → Pong Buffer 0xD1100504/0xD2800504 [2] = 0 → Ping Buffer	
[5]	EP9_OVLD	R	Current Endpoint 9 OUT Buffer Valid Flag Status Which buffer is the cur_vld_buffer depends on 0xD1100504/0xD2800504 [2]. 0xD1100504/0xD2800504 [2] = 1 → Ping Buffer 0xD1100504/0xD2800504 [2] = 0 → Pong Buffer	
[4]	EP8_IVLD	R	Current Endpoint 8 IN Buffer Valid Flag Status Which buffer is the cur_buffer depends on 0xD1100504/0xD2800504 [2]. 0xD1100504/0xD2800504 [2] = 1 → Ping Buffer 0xD1100504/0xD2800504 [2] = 0 → Pong Buffer	
[3]	-	-	Reserved	-
[2]	Current_Buffer	R	Current Buffer Selection	0: Buffer1 (PONG) buffer is controllable for CPU/DMA. 1: Buffer0 (PING) buffer is controllable for CPU/DMA.

Bit	Function	Type	Description	Condition
[1]	Switch_Buffer	W	Write 1 to switch between Buffer 0 and Buffer 1.	0: Use the same buffer as the current one. 1: Use the other buffer, not the current one.
[0]	Auto_Switch_En	R/W		0: PING PONG will NOT auto switch. 1: PING PONG will auto switch.

EP89FS
0xD1100508/0xD2800508
Endpoint 8 and Endpoint 9 Status

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-	A_FIFO_E MPTY	A_EP9_OV LD	A_EP8_IVL D	-	FIFO_EMPTY	EP9_OVLD	EP8_IVLD
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:7]	-	-	Reserved	-
[6]	A_FIFO_EMPTY	R	Another Buffer Status	0: Not Empty. 1: Empty.
[5]	A_EP9_OVLD	R	Another Endpoint 9 OUT Buffer Valid Flag Status The function is the same as 0xD1100504/0xD2800504[7].	0: Not ready to receive data. 1: Ready to receive data.
[4]	A_EP8_IVLD	R	Another Endpoint 8 IN Buffer Valid Flag Status The function is the same as 0x504[6].	0: Invalid (Data has been read by the host). 1: Valid (Data is still in Buffer).

Bit	Function	Type	Description	Condition
[3]	-	-	Reserved	-
[2]	FIFO_EMPTY	R	Data in Current Buffer Status	0: Not Empty. 1: Empty.
[1]	EP9_OVLD	R	The function is the same as 0xD1100504/0xD2800504[5].	
[0]	EP8_IVLD	R	Current Endpoint 8 IN Buffer Valid Flag Status The function is the same as 0xD1100504/0xD2800504[4].	0: Invalid (Data has been read by the host). 1: Valid (Data is still in Buffer).

EP89PICL
0xD110050C/0xD280050C
Endpoint 8 and Endpoint 9 PING Data Count
Low Byte

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EP89_PING_CNTRL[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:8]	-	-	Reserved	-
[7:0]	EP89_PING_CNTRL[7:0]	R/W	Write: Modify read or write pointer. Read: BULK_OUT = Valid Byte Count BULK_IN before SET_IN_VLD = Write pointer. BULK_IN after SET_IN_VLD = Valid Byte Count	

EP89PICH
0xD1100510/0xD2800510
Endpoint 8 and Endpoint 9 PING Data Count
High Byte

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-					Reset_PING_CNTR	EP89_PING_CNTRH[1:0]	
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:3]	-	-	Reserved	-
[2]	Reset_PING_CNTR	R/W	Write 1: BULK_OUT = Move r_ptr to 0. BULK_IN = Move w_ptr to 0.	
[1:0]	EP89_PING_CNTR RH[1:0]	R/W	Write: Modify read or write pointer. Read: BULK_OUT = Valid Byte Count BULK_IN before SET_IN_VLD = Write pointer. BULK_IN after SET_IN_VLD = Valid Byte Count	

EP89POCL
0xD1100514/0xD2800514
Endpoint 8 and Endpoint 9 PONG Data Count
Low Byte

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EP89_PONG_CNTRL[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:8]	-	-	Reserved	-
[7:0]	EP89_PONG_CNTRL[7:0]	R/W	Write: Modify read or write pointer Read: BULK_OUT = Valid Byte Count BULK_IN before SET_IN_VLD = Write pointer. BULK_IN after SET_IN_VLD = Valid Byte Count	

EP89POCH
0xD1100518/0xD2800518
Endpoint 8 and Endpoint 9 PONG Data Count
High Byte

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-					ResetPONG_CNTR	EP89_PONG_CNTRH[1:0]	
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:3]	-	-	Reserved	-
[2]	ResetPONG_CNT R	R/W	Write 1: BULK_OUT = Move r_ptr to 0. BULK_IN = Move w_ptr to 0.	
[1:0]	EP89_PONG_CNT RH[10:]	R/W	Write: Modify read or write pointer Read: BULK_OUT = Valid Byte Count BULK_IN before SET_IN_VLD = Write pointer. BULK_IN after SET_IN_VLD = Valid Byte Count	

EP89FDP 0xD110051C/0xD280051C Endpoint 8 and Endpoint 9 Data Port

Bit	31	30	29	28	27	26	25	24
Function	EP89_DATA_PORT[31:24]							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	EP89_DATA_PORT[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	EP89_DATA_PORT[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EP89_DATA_PORT[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:0]	EP89_DATA_PORT[31:0]	R/W	USB Device Endpoint 8 and Endpoint 9 FIFO Data Port For how many valid reading or writing byte of data is valid, please refer to EP89VB (0xD1100520/0xD28000520).	

EP89VB 0xD1100520/0xD2800520 Endpoint 8 and Endpoint 9 Data Port Valid Byte

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-				EP89VB[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:4]	-	-	Reserved	-
[3:0]	EP89VB[3:0]	R/W	USB Device Endpoint 8 and Endpoint 9 Data Port Valid Byte EP89VB indicates which Byte(s) is valid in EP89DATA (0xD110051C/0xD280051C). Data Port [31:0] = {Byte3, Byte2, Byte1, Byte0}.	EP89VB = 0: No byte is valid. EP89VB[0] = 1: Byte0 is valid. EP89VB[1] = 1: Byte1 is valid. EP89VB[2] = 1: Byte2 is valid. EP89VB[3] = 1: Byte3 is valid.

EP8INAKCN 0xD1100524/0xD2800524 Endpoint 8 IN NAK Count

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-				EP8_INAK_CNT[13:8]			
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EP8_INAK_CNT[7:0]/EP8_INAK_THRESHOLD[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:14]	-	-	Reserved	-
[13:8]	EP8_INAK_CNT [13:8]	R	Read EP8_INAK_CNT and user can obtain the value of EP8_INAK_CNT[13:0]. EP8_INAK_CNT is 14bits. When EP8_INAK_ICNT[13:6] == EP8_INAK_THRESHOLD[7:0], UDC_NEW_IF[2] (0xD1100420/0xD2800420) will be triggered.	
[7:0]	EP8_INAK_CNT [7:0]/EP8_INAK_THRESHOLD[7:0]	R/W	Write a suitable value to EP8_INAK_THRESHOLD to determine the interrupt will happen after how many NAK counts. For reading EP8_INAK_CNT[7:0], please refer to the description in EP8_INAK_CNT[13:8].	

EP9ONAKCN
0xD1100528/0xD2800528
Endpoint 9 OUT NAK Count

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-		EP9_ONAK_CNT[13:8]					
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EP9_ONAK_CNT[7:0]/EP9_ONAK_THRESHOLD[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:14]	-	-	Reserved	-
[13:8]	EP9_ONAK_CNT [13:8]	R	Read EP9_ONAK_CNT and user can obtain the value of EP9_ONAK_CNT[13:0]. EP9_ONAK_CNT is 14bits. When EP9_ONAK_ICNT[13:6] == EP9_ONAK_THRESHOLD[7:0], UDC_NEW_IF[3] (0xD1100420/0xD2800420) will be triggered.	
[7:0]	EP9_ONAK_CNT [7:0]/EP9_ONAK_THRESHOLD[7:0]	R/W	Write a suitable value to EP9_ONAK_THRESHOLD to determine the interrupt will happen after how many NAK counts. For reading EP9_ONAK_CNT[7:0], please refer to the description in EP9_ONAK_CNT[13:8].	

UDEP89_INTF		0xD110052C/0xD280052C				Endpoint 8 and Endpoint 9 Interface Setting		
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EP89ALT_SET[3:0]				EP89INTF_SET[3:0]			
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:8]	-	-	Reserved	-
[7:4]	EP89ALT_SET[3:0]	R/W	Endpoint 8 and Endpoint 9 Alternate Setting	
[3:0]	EP89INTF_SET[3:0]	R/W	Endpoint 8 and Endpoint 9 Interface Number	

EP CER		0xD1100580/0xD2800580				Endpoint 12 Event		
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-			EPC_DMA_Finish	EPC_NAK	EPC_OUTNAK	EPC_ER	EPC_EN
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:5]	-	-	Reserved	-
[4]	EPC_DMA_Finish	R/W	EPC_DMA_Finish will be set as 1 when DMA finishes. Write 1 and this bit will be cleared.	
[3]	EPC_NAK	R/W	When a transaction is sent by the host, and EPC_EN = 0, this bit will be set as 1. Write 1 and this bit will be cleared.	
[2]	EPC_OUTNAK	R/W	When the Out transaction fails, and EPC_EN = 1, this bit will be set as 1. Write 1 and this bit will be cleared.	
[1]	EPC_ER	R/W	Endpoint 12 Isochronous Out Package Ready This bit will be set as 1 if an OUT Package is loaded into isochronous out FIFO. Write 1 and this bit will be cleared automatically. In DMA mode, EPC_ER will be cleared automatically when MCU DMA is done.	
[0]	EPC_EN	R/W	Endpoint 12 Isochronous Out Enable Writing 1 will enable incoming packet for OUT Data. In DMA mode, this bit will be cleared automatically when USB transfer finishes. Writing 0 will clear this bit.	

EPCCTRL 0xD1100584/0xD2800584 Endpoint12 Control

Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	-				RWPOINT_ CLEAR	-		AISO_OUT _DMAEN
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:4]	-	-	Reserved	-
[3]	RWPOINT_CLEAR	R/W	Isochronous OUT Read/Write Point Clear	0: No action. 1: R/W Point is cleared.
[2:1]	-	-	Reserved	-
[0]	AISO_OUT_DMA EN	R/W	Audioe Isochronous OUT DMA Enable	0: No action. 1: Audioe isochronous OUT DMA is enabled.

EPCWCNT					0xD1100588/0xD2800588				Endpoint 12 Write Counter	
Bit	31	30	29	28	27	26	25	24		
Function	-									
Default	0	0	0	0	0	0	0	0	0	

Bit	23	22	21	20	19	18	17	16
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	-							IOUTWC[8]
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	IOUTWC[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:9]	-	-	Reserved	-
[8:0]	IOUTWC[8:0]	R	Endpoint 12 Isochronous Out Write Counter Read IOUTWC and user will obtain the data number in the isochronous Out FIFO.	

EPCFR		0xD110058C/0xD280058C					Endpoint 12 Function	
Bit	31	30	29	28	27	26	25	24
Function	-							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	-			EPC_RP[8:4]				
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	EPC_RP[3:0]				-			EPC_WP[8]
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	EPC_WP[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:21]	-	-	Reserved	-
[20:12]	EPC_RP[8:0]	R	Endpoint 12 Read Point.	
[11:9]	-	-	Reserved	-
[8:0]	EPC_WP[8:0]	R	Endpoint 12 Write Point EPC_WP will be set as 0 when USB reception successes.	

EPC_DATA		0xD1100590/0xD2800590					Endpoint 12 FIFO Data Port	
Bit	31	30	29	28	27	26	25	24
Function	UDEPCDATA[31:24]							
Default	0	0	0	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	UDEPCDATA[23:16]							
Default	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8
Function	UDEPCDATA[15:8]							
Default	0	0	0	0	0	0	0	0

Bit	7	6	5	4	3	2	1	0
Function	UDEPCDATA[7:0]							
Default	0	0	0	0	0	0	0	0

Bit	Function	Type	Description	Condition
[31:0]	UDEPCDATA[31:0]	R/W	USB Device EndpointC FIFO Data Port	

IP_NAME		0xD1100FF8/0xD2800FF8					USB Device IP Name	
Bit	31	30	29	28	27	26	25	24
Function	IP_NAME[31:24]							
Default	0	1	0	1	0	0	1	1

Bit	23	22	21	20	19	18	17	16
Function	IP_NAME[23:16]							
Default	0	0	1	1	0	0	1	0

Bit	15	14	13	12	11	10	9	8
Function	IP_NAME[15:8]							
Default	0	0	1	1	0	0	1	0

Bit	7	6	5	4	3	2	1	0
Function	IP_NAME[7:0]							
Default	0	0	1	1	0	0	1	0

Bit	Function	Type	Description	Condition
[31:0]	IP_NAME[31:0]	R	The USB Device IP Name	

VERSION		0xD1100FFC/0xD2800FFC					USB Device IP Version	
Bit	31	30	29	28	27	26	25	24
Function	VERSION[31:24]							
Default	0	0	1	0	0	0	0	0

Bit	23	22	21	20	19	18	17	16
Function	VERSION[23:16]							
Default	0	0	0	1	0	0	1	0

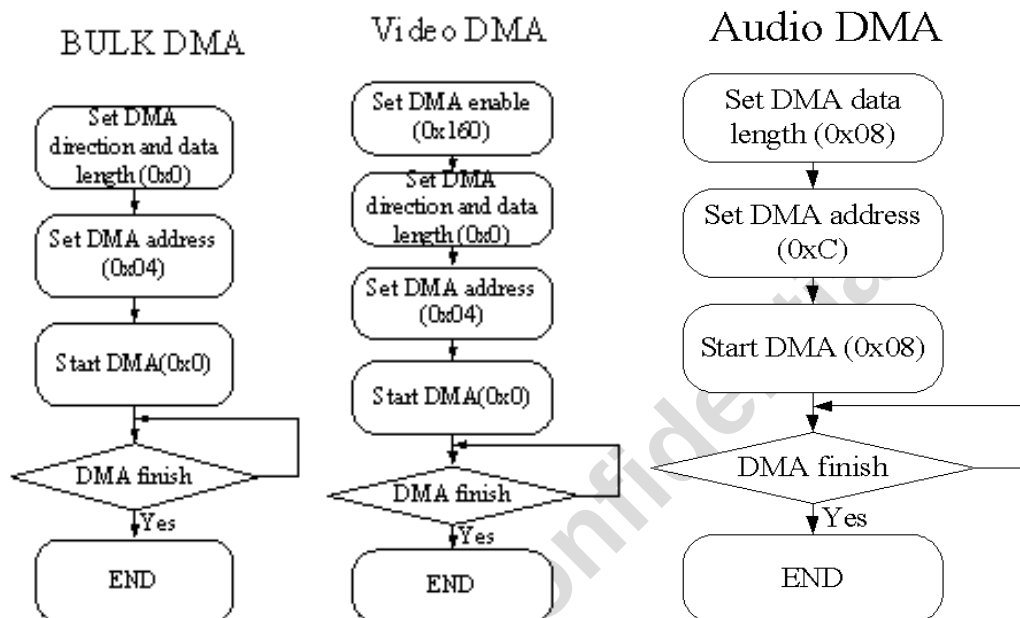
Bit	15	14	13	12	11	10	9	8
Function	VERSION[15:8]							
Default	0	0	0	1	0	0	1	0

Bit	7	6	5	4	3	2	1	0
Function	VERSION[7:0]							
Default	0	0	0	1	0	1	0	0

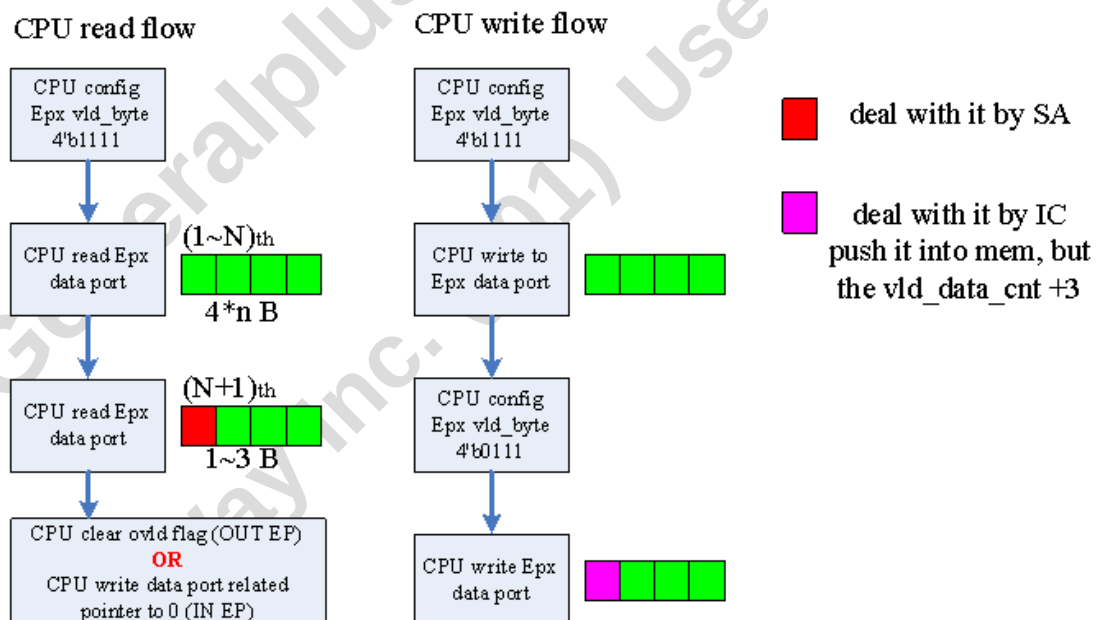
Bit	Function	Type	Description	Condition
[31:0]	VERSION[31:0]	R	The Date of Device Version	

1.7. DMA and PIO Flow

1.7.1. DMA Flow



1.7.2. PIO Flow



For example, there are 21 bytes in EP0 data port, and user wants to copy all of them to the buffer. Since $21/4 = 5$, set vld byte[3:0] = 0x0F, and copy the data from EP0 data port to the specified buffer. Repeat this action for 5 times.

Since $21\%4 = 1$, which means the the remainder is 1, set `vald_byte[3:0] = 0x01`, and copy the last one byte from EP0 data port to the buffer to end this data copy process. The following table shows the relationship between byte number to get and the value in `vald_byte` register.

Byte Number to Get	Vald_byte Value
1	0x01 (b'0001)
2	0x03 (b'0011)
3	0x07 (b'0111)
4	0x0F (b'1111)

2. Control Flow for USB Transaction

2.1. USB Transactions

USB supports four types of transfer for data transfer, they are:

- Control Transfer
- Bulk Transfer
- Interrupt Transfer
- Isochronous (ISO) Transfer

Control transfer is the most important transfer and this function must exist in every USB device. It contains three transaction stages: Setup, Data and Status. The USB host can use control transfer to send standard command and know what type of the current USB device is. The host can obtain the device information through control transfer, and it will apply the functions (MSDC, UVC, UAC, and so on) of the device according to the information.

Bulk transfer has one transaction type which is the same as Interrupt. The difference is that Bulk is used to transmit large data, so its maximum data size is larger than Interrupt. (i.e H.S 512 bytes, F.S 64 bytes)

Isochronous transfer has one transaction type, and the biggest difference with the others transactions is that there's no handshake stage during isochronous transaction. Thus this transfer is used to transfer real-time data such as video or audio stream.

In this chapter, it will show user the USB device register combination flow, and how to implement the transactions of control, bulk, interrupt and isochronous transfer.

2.2. Control Transfer

2.2.1. Setup Stage

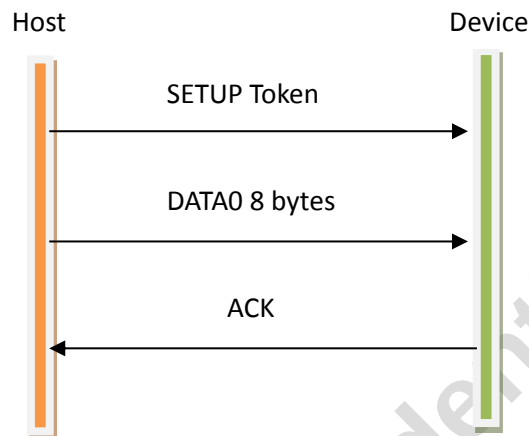


Figure 2.1 EP0 SETUP

Control flow of register combination for **PIO mode**:

1. Wait for 0xD1100400/0xD2800400[2] to get a SETUP command from the host.
2. If this bit = 1, the device will get SETUP data from EP0 data port 0xD1100348/0xD2800348. Read the register twice, and user will get data with 8 bytes length.

2.2.2. Data Stage

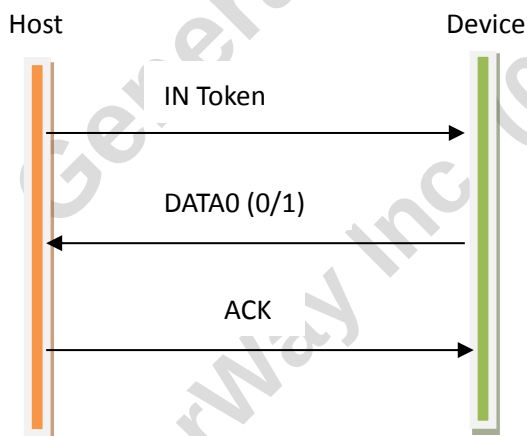


Figure 2.2 EP0 DATA IN

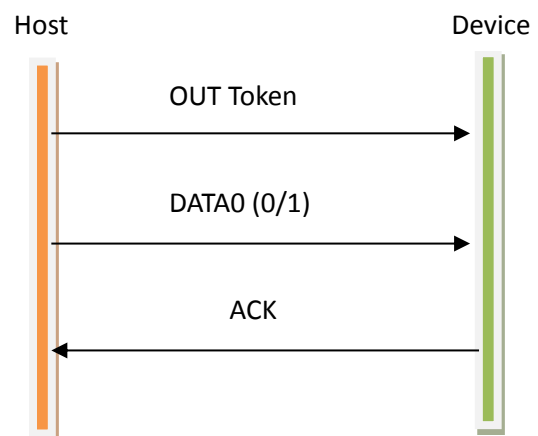


Figure 2.3 EP0 DATA OUT

Control flow of register combination for **DATA IN PIO mode**:

1. Write 1 to 0xD110034C/0xD280034C, and set control IN.
2. Write 0 to 0xD1100350/0xD2800350, and reset EP0 data count.

3. 0xD1100354/0xD2800354 EP0 data port will co-work with 0xD1100340/0xD2800340 data valid byte to write data into EP0 FIFO. Please refer to section 1.7 PIO flow.
4. User can set data length in 0xD1100350/0xD2800350.
5. Write 1 to 0xD1100400/0xD2800400 [4], and clear EP0 IN transaction flag.
6. Write 1 to 0xD1100404/0xD2800404 [4], and enable EP0 IN transaction interrupt flag.
7. Write 1 to 0xD110034C/0xD280034C [3], and EP0 Data IN will be valid.

Control flow of register combination for **DATA OUT PIO mode**:

1. Write 1 to 0xD1100400/0xD2800400 [3], and clear EP0 OUT traction flag.
2. Write 1 to 0xD1100404/0xD2800404 [3], and enable EP0 OUT transaction interrupt flag.
3. Write 1 to 0xD110034C/0xD280034C [0], and set control OUT.
4. Wait for interrupt of EP0 OUT.
5. Check 0xD110034C/0xD280034C [6] to see whether EP0 OUT buffer is valid.
6. Get data count from 0xD1100350/0xD2800350.
7. Get EP0 OUT data from EP0 data port 0xD1100354/0xD2800354 and valid byte 0xD1100340/0xD2800340. Please refer to PIO flow.
8. Write 1 to 0xD110034C and 0xD280034C [5], and clear EP0 data OUT valid and free buffer.

2.2.3. Status Stage

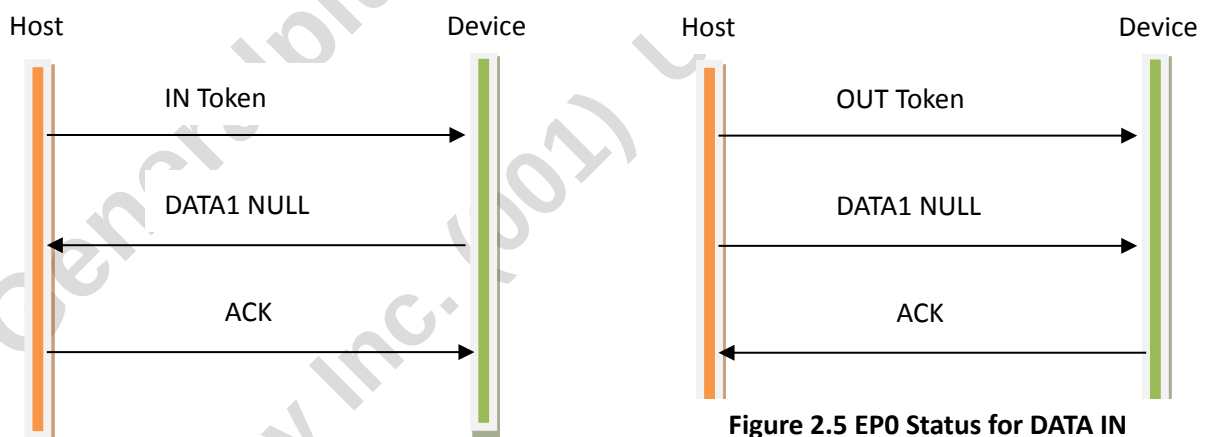


Figure 2.4 EP0 Status for DATA OUT/NO DATA

Control flow of register combination for **sending null DATA IN**:

1. Write 1 to 0xD110034C/0xD280034C [0], and set control IN.
2. Write 1 to 0xD1100400/0xD2800400 [4], and clear EP0 in transaction flag.
3. Write 1 to 0xD1100404/0xD2800404 [4], and enable EP0 IN transaction interrupt flag.
4. Write 1 to 0xD110034C/0xD280034C, and EP0 Data IN will be valid.
5. Get interrupt from the USB device.
6. If 0xD110034C/0xD280034C [4] = 0, see whether the data has been received by the host.

2.3. Bulk Transfer

Bulk Transaction

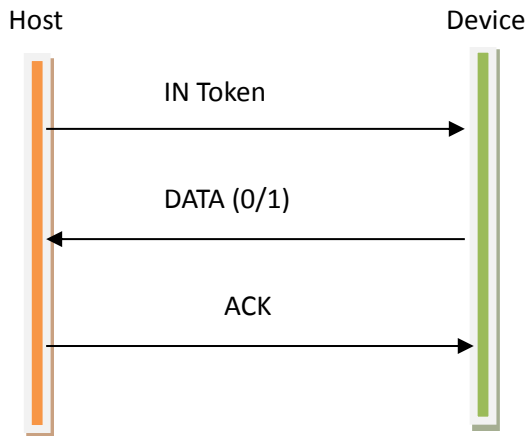


Figure 2.6 Bulk DATA IN

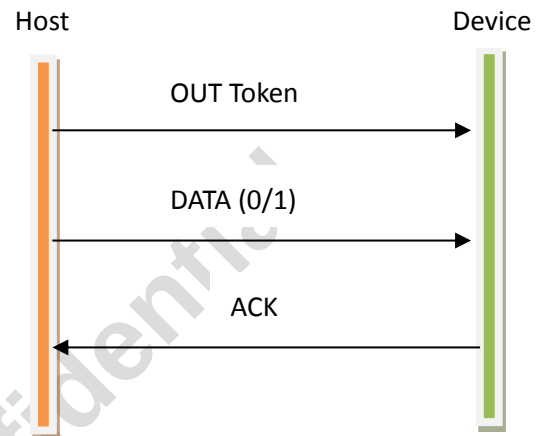


Figure 2.7 Bulk DATA OUT

Control flow of register combination for **DATA IN PIO mode** (size < 512 bytes):

Endpoint 1 – Auto Switch (0xD1100334/0xD2800334 [0] = 1) for ping/pong FIFO

1. Write 1 to 0xD1100330/0xD2800330 [2], and reset FIFO.
2. Write 1 to 0xD1100330/0xD2800330 [8], and enable BULK IN.
3. Send data to 0xD1100370/0xD2800370 and co-work with 0xD1100398/0xD2800398. Please refer to PIO flow.
4. Write 1 to 0xD1100404/0xD2800404 [6], and enable endpoint 1 IN transaction interrupt. Write 1 to 0xD1100400/0xD2800400 [6], and clear endpoint 1 IN interrupt flag.
5. Write 1 to 0xD1100330/0xD2800330 [4], and set endpoint 1 IN as valid.
6. Wait for 0xD1100400/0xD2800400 [6] endpoint 1 transaction interrupt.
7. Disable endpoint 1 IN transaction interrupt 0xD1100404/0xD2800404 [6].

Note: After ep1_usb_succ IRQ occurs (the host received the last data), EP1_IN_ENA will be cleared automatically.

Endpoint 1 – Manual Switch (0xD1100334/0xD2800334 [0] = 0)

1. Write 1 to 0xD1100330/0xD2800330 [2], and reset FIFO.
2. Write 1 to 0xD1100330/0xD2800330 [8], and enable BULK IN.
3. Send data to 0xD1100370/0xD2800370, and 0xD1100370/0xD2800370 will co-work with 0xD1100398/0xD2800398. Please refer to section 1.6 PIO flow.
4. Write 1 to 0xD1100404/0xD2800404 [6], and enable endpoint 1 IN transaction interrupt. Write 1 to 0xD1100400/0xD2800400 [6], and clear endpoint 1 IN interrupt flag.
5. Write 1 to 0xD1100330/0xD2800330 [4], and set endpoint 1 IN as valid.

6. Write 1 to 0xD1100334/0xD2800334 [1], and switch PIPO buffer.
7. Wait for 0xD1100400/0xD2800400 [6] endpoint 1 transaction interrupt.
8. Disable endpoint 1 IN transaction interrupt 0xD1100404/0xD2800404 [6].

Note: After ep1_usb_succ IRQ occurs (the host received the last data), EP1_IN_ENA will be cleared automatically.

Endpoint 8 – Auto Switch (0xD1100504/0xD2800504 [0] = 1)

1. Write 1 to 0xD1100500/0xD2800500 [2], and reset FIFO.
2. Write 1 to 0xD1100500/0xD2800500 [7], and enable BULK IN.
3. Send data to 0xD110051C/0xD280051C, and 0xD110051C/0xD280051C will co-work with 0xD1100520/0xD2800520. Please refer to section 1.6 PIO flow.
4. Write 1 to 0xD1100420/0xD2800420 [4], and enable endpoint 8 IN transaction interrupt. Write 1 to 0xD1100424/0xD2800424 [4], and clear endpoint 8 IN interrupt flag.
5. Write 1 to 0xD1100500/0xD2800500 [4], and set endpoint 8 IN as valid.
6. Wait for 0xD1100420/0xD2800420 [4] endpoint 8 transaction interrupt.
7. Disable endpoint 8 IN transaction interrupt 0xD1100424/0xD2800424 [4].

Note: After ep8_usb_succ IRQ occurs (the host received the last data), EP8_IN_ENA will be cleared automatically.

Control flow of register combination for **DATA IN DMA mode** (size = multiple of 512 bytes):

Endpoint 1

1. Write 1 to 0xD1100330/0xD2800330 [2], and reset FIFO.
2. Write 1 to 0xD1100330/0xD2800330 [8], and enable BULK IN.
3. Write 1 to 0xD1100334/0xD2800334 [0], and enable endpoint 1 and endpoint 2 PIPO FIFO auto switch.
4. User can set DMA data address in 0xD1100004/0xD2800004 [0:31].
5. 0xD1100000/0xD2800000 [0:21] = DMA data length.
6. Write 0 to 0xD1100000/0xD2800000 [28] to determine BULK IN DMA direction.
7. Write 1 to 0xD1100400/0xD2800400 [15], and clear this bit.
8. Write 1 to 0xD1100404/0xD2800404 [15], and enable endpoint 1 and endpoint 2 DMA IN data finish interrupt.
9. Write 1 to 0xD1100000/0xD2800000 [31], and enable DMA.
10. Wait for 0xD1100400/0xD2800400 [15] = 1 and then disable 0xD1100404/0xD2800404 [15] endpoint 1 DMA IN data finish interrupt enable.

Note: After ep1_usb_succ IRQ occurs 8 times (the host received the last data), EP1_IN_ENA will be cleared automatically. (Transfer Data Length = 4K)

Endpoint 8

1. Write 1 to 0xD1100500/0xD2800500 [2], and reset FIFO.
2. Write 1 to 0xD1100500/0xD2800500 [7], and enable BULK IN.
3. Write 1 to 0xD1100504/0xD2800504 [0], and enable endpoint 8 and endpoint 9 PIPO FIFO autoswitch.
4. 0xD1100010/0xD2800010 [0:21] set data length.
5. 0xD1100014/0xD2800014 set buffer address.
6. 0xD1100010/0xD2800010 [28] write 0 to set direction IN.
7. Write 1 to 0xD1100420/0xD2800420 [0], and clear flag.
8. Write 1 to 0xD1100424/0xD2800424 [0], and enable endpoint 8 and endpoint 9 DMA interrupt.
9. Write 1 to 0xD1100010/0xD2800010 [31], and enable DMA.
10. Wait for 0xD1100420/0xD2800420 [0] = 1 and then disable 0xD1100424/0xD2800424 [0] endpoint 8 and endpoint 9 DMA IN data finish interrupt.

Note: After ep8_usb_succ IRQ occurs 8 times (the host received the last data), EP8_IN_ENA will be cleared automatically. (if transfer data length = 4K)

Control flow of register combination for **DATA OUT PIO mode** (size < 512 bytes):

Endpoint 2 – Auto Switch (0xD1100334/0xD2800334 [0] = 1)

1. Write 1 to 0xD1100330/0xD2800330 [2], and reset FIFO.
2. Write 1 to 0xD1100330/0xD2800330 [9], and enable BULK OUT.
3. Write 1 to 0xD1100400/0xD2800400 [7] and clear 0xD1100400/0xD2800400 [7].
4. Write 1 to 0xD1100404/0xD2800404 [7], and enable endpoint 2 OUT transaction interrupt.
5. Wait for 0xD1100400/0xD2800400 [7] = 1.
6. Write 0 to 0xD1100404/0xD2800404 [7] and disable 0xD1100404/0xD2800404 [7].
7. Check 0xD1100334/0xD2800334 [2] to know current buffer is PING or PONG.
8. Read out data count from 0xD1100368/0xD2800368 (0xD110039C/0xD280039C) for low byte and 0xD110036C/0xD280036C (0xD11003A0/0xD28003A0) for high byte.
9. Read out data from 0xD1100370/0xD2800370 endpoint 1 and endpoint 2 data port. Collect and store the data in the buffer after referring to valid bit field. Please refer to section 1.6 PIO flow.
10. Get data from endpoint 1 and endpoint 2 FIFO done then 0xD1100330/0xD2800330 [3] write 1 to clear endpoint 2 out buffer valid flag.

Note: After ep2_usb_succ IRQ occurs (the host transferred the last data), EP2_OUT_ENA will be cleared automatically.

Endpoint 2 – Manual Switch (0xD1100334/0xD2800334 [0] = 0)

1. Write 1 to 0xD1100330/0xD2800330 [2], and reset FIFO.
2. Write 1 to 0xD1100330/0xD2800330 [9], and enable BULK OUT.
3. 0xD1100400/0xD2800400 [7] write 1 clear.
4. Write 1 to 0xD1100404/0xD2800404 [7], and enable endpoint 2 OUT transaction interrupt.
5. Wait for 0xD1100400/0xD2800400 [7] = 1.
6. Disable 0xD1100404/0xD2800404 [7], write 0.
7. Check 0xD1100334/0xD2800334 [7] is valid or not. If it's valid, write 1 to 0xD1100334/0xD2800334 [1], and switch PIFO buffer to get correct bulk out data.
8. Check 0xD1100334/0xD2800334 [2] to know the current buffer is PING or PONG.
9. Get data count from 0xD1100368/0xD2800368 (0xD110039C/0xD280039C) for low byte and 0xD110036C/0xD280036C (0xD11003A0/0xD28003A0) for high byte.
10. Get data from 0xD1100370/0xD2800370 endpoint 1 and endpoint 2 data port (set valid bit field and write into buffer). Please refer to section 1.6 PIO flow.
11. Get data from endpoint 1 and endpoint 2 FIFO done then 0xD1100330/0xD2800330 [3] write 1 to clear endpoint 2 out buffer valid flag.

Note: After ep2_usb_succ IRQ occurs (the host transferred the last data), EP2_OUT_ENA will be cleared automatically.

Endpoint 9

1. Write 1 to 0xD1100500/0xD2800500 [2], and reset FIFO.
2. Write 1 to 0xD1100500/0xD2800500 [8], and enable BULK OUT.
3. Write 1 to 0xD1100420/0xD2800420 [5], and clear endpoint 9 OUT transaction flag.
4. Write 1 to 0xD1100424/0xD2800424 [5], and enable endpoint 9 OUT transaction interrupt.
5. Wait 0xD1100420/0xD2800420 [5] = 1.
6. Disable 0xD1100424/0xD2800424 [5], write 0.
7. Check 0xD1100504/0xD2800504 [2] to know current buffer is PING or PONG.
8. Get data count from 0xD110050C/0xD280050C (0xD1100514/0xD2800514) for low byte and 0xD1100510/0xD2800510 (0xD1100518/0xD2800518) for high byte.
9. Get data from 0xD110051C/0xD280051C endpoint 8 and endpoint 9 data port (set valid bit field and write into buffer). Please refer to section 1.6 PIO flow.
10. Get data from endpoint 1 and endpoint 2 FIFO done then 0xD1100500/0xD2800500 [3] write 1 to clear endpoint 9 out buffer valid flag.

Note: After ep9_usb_succ IRQ occurs (the host transferred the last data), EP9_OUT_ENA will be cleared automatically.

Control flow of register combination for **DATA OUT DMA mode** (size = multiple of 512 bytes):

Endpoint 2

1. Write 1 to 0xD1100330/0xD2800330 [2], and reset FIFO.
2. Write 1 to 0xD1100334/0xD2800334 [0], and enable endpoint 1 and endpoint 2 PIPO FIFO auto switch.
3. Write 1 to 0xD1100330/0xD2800330 [9], and enable BULK OUT.
4. 0xD1100000/0xD2800000 [0:21] set data length.
5. 0xD1100004/0xD2800004 set buffer address.
6. 0xD1100000/0xD2800000 [28] write 1 to set direction.
7. 0xD1100088/0xD2800088 [25] write 1 to clear flag.
8. Write 1 to 0xD1100084/0xD2800084 [25], and enable endpoint 1 and endpoint 2 DMA interrupt.
9. Write 1 to 0xD1100000/0xD2800000 [31], and enable DMA.
10. Wait for 0xD1100688/0xD2800688 [25] = 1 and check buffer.

Note: After ep2_usb_succ IRQ occurs (the host transferred the last data), EP2_OUT_ENA will be cleared automatically.

Endpoint 9

1. Write 1 to 0xD1100500/0xD2800500 [2], and reset FIFO.
2. Write 1 to 0xD1100504/0xD2800504 [0], and enable endpoint 8 and endpoint 9 PIPO FIFO auto switch.
3. Write 1 to 0xD1100500/0xD2800500 [8], and enable BULK OUT.
4. 0xD1100010/0xD2800010 [0:21] set data length.
5. 0xD1100014/0xD2800014 set buffer address.
6. 0xD1100010/0xD2800010 [28] write 1 to set direction OUT.
7. 0xD1100088/0xD2800088 [29] write 1 to clear flag.
8. Write 1 to 0xD1100084/0xD2800084 [29], and enable endpoint 8 and endpoint 9 DMA interrupt.
9. Write 1 to 0xD1100010/0xD2800010 [31], and enable DMA.
10. Wait 0xD1100688/0xD2800688 [29] and check buffer.

Note: After ep9_usb_succ IRQ occurs (the host transferred the last data), EP9_OUT_ENA will be cleared automatically.

2.4. Interrupt Transfer

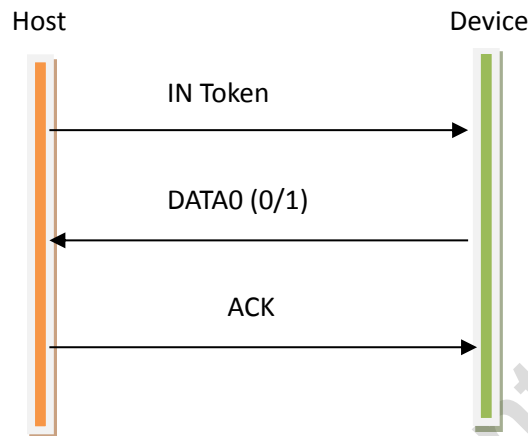


Figure 2.8 INT IN

Control flow of register combination for **DATA IN PIO mode**:

Endpoint 3

1. Send data to endpoint 3 data port 0xD110037C/0xD280037C, and 0xD110037C/0xD280037C will co-work with 0xD1100380/0xD2800380 data valid byte register. Please refer to section 1.7 PIO flow.
2. Write 1 to 0xD1100400/0xD2800400 [14], and clear endpoint 3 IN interrupt flag.
3. Write 1 to 0xD1100404/0xD2800404 [14], and enable endpoint 3 IN transaction interrupt
4. Write 1 to 0xD1100374/0xD2800374 [0], and enable endpoint 3 IN.
5. Write 1 to 0xD1100374/0xD2800374 [3], and set endpoint 3 as valid bit.
6. Wait for 0xD1100400/0xD2800400 [14] = 1. Repeat 1~5 if there's still data which has to be sent to the host.

Endpoint 4

1. Send data to endpoint 4 data port 0xD1100108/0xD2800108, and 0xD1100108/0xD2800108 will co-work with 0xD110010C/0xD280010C data valid byte register. Please refer to section 1.6 PIO flow.
2. Write 1 to 0xD1100400/0xD2800400 [20], and clear endpoint 4 IN interrupt flag.
3. Write 1 to 0xD1100404/0xD2800404 [20], and enable endpoint 4 IN transaction interrupt.
4. Write 1 to 0xD1100100/0xD2800100 [0], and enable endpoint 4 IN.
5. Write 1 to 0xD1100100/0xD2800100 [3], and set endpoint 4 as valid bit.
6. Wait for 0xD1100400/0xD2800400 [20] = 1. Repeat 1~5 if there's still data which has to be sent to the host.

Endpoint 6

1. Send data to endpoint 6 data port 0xD1100188/0xD2800188 and, 0xD1100188/0xD2800188 will co-work with 0xD110018C/0xD280018C data valid byte register. Please refer to section 1.6 PIO flow.
2. Write 1 to 0xD1100400/0xD2800400 [22], and clear endpoint 6 IN interrupt flag.
3. Write 1 to 0xD1100404/0xD2800404 [22], and enable endpoint 6 IN transaction interrupt.
4. Write 1 to 0xD1100180/0xD2800180 [0], and enable endpoint 6 IN.
5. Write 1 to 0xD1100180/0xD2800180 [3], and set endpoint 6 as valid bit.
6. Wait for 0xD1100400/0xD2800400 [22] = 1. Repeat 1~5 if there's still data which has to be sent to the host.

2.5. Isochronous Transfer

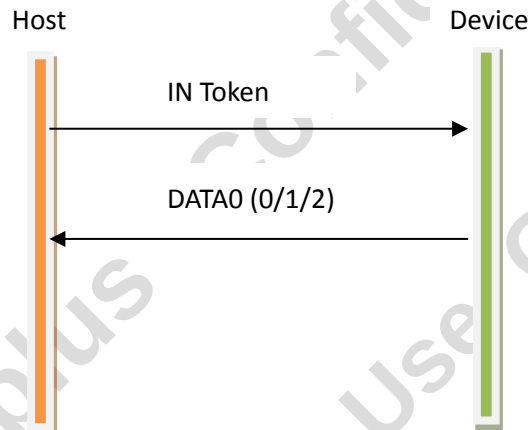


Figure 2.9 Isochronous IN

Control flow of register combination for **DATA IN DMA mode**:

Endpoint 5

1. Make sure 0xD1100150/0xD2800150 [0] = 1 to enable endpoint 5 endpoint.
2. Write 1 to 0xD1100160/0xD2800160 [0], and enable endpoint 5 DMA function.
3. Write 1 to 0xD1100140/0xD2800140 [1], and flush endpoint 5 FIFO
4. Write 0 to 0xD1100140/0xD2800140 [1], and stop flushing endpoint 5 FIFO.
5. Write 1 to 0xD1100000/0xD2800000 [29], and flush endpoint 1 and endpoint 2 DMA controller.
6. Wait 0xD1100000/0xD2800000 [30] = 1 to finish flushing endpoint 1 and endpoint 2 DMA controller.
7. 0xD1100004/0xD2800004 [0:31] set DMA data address.
8. 0xD1100000/0xD2800000 [0:21] = DMA data length.
9. Write 0 to 0xD1100000/0xD2800000 [28], and it means BULK IN DMA direction is reading data from the system.

10. Write 1 to 0xD1100400/0xD2800400 [24], and clear DMA IN finish interrupt flag.
11. Write 1 to 0xD1100404/0xD2800404 [24], and enable DMA IN finish interrupt.
12. Write 1 to 0xD1100000/0xD2800000 [31], and enable endpoint 1 and endpoint 2 DMA.
13. Wait for 0xD1100400/0xD2800400 [24] = 1 to know whether the data in isochronous buffer is sent by the device.

Endpoint 7

1. Write 1 to 0xD11001C0/0xD28001C0 [3], 0xD11001C0/0xD28001C0 [0], and set endpoint 7 IN as valid.
2. Write 1 to 0xD11001C0/0xD28001C0 [7], and flush endpoint 7 FIFO.
3. Write 0 to 0xD11001C0/0xD28001C0 [7], and stop flushing endpoint 7 FIFO.
4. 0xD110000C/0xD280000C set DMA data address.
5. 0xD1100008/0xD2800008 [15:0] set DMA data byte count.
6. Write 1 to 0xD1100008/0xD2800008 [31], and enable endpoint 7 DMA.
7. Write 1 to 0xD1100400/0xD2800400 [23], and clear flag.
8. Write 1 to 0xD1100404/0xD2800404 [23], and enable endpoint 7 IN transaction flag.
9. Wait for 0xD1100400/0xD2800400 [23] = 1 to see whether endpoint 7 IN transaction finishes. After the transaction finishes, disable 0xD1100404/0xD2800404 [23] interrupt.